

THE RURAL MAGAZINE

AND
LITERARY EVENING
FIRE-SIDE



Vol. 1 No. 7

THE RURAL MAGAZINE

AND
LITERARY EVENING
FIRE-SIDE



Vol. 1 No. 7

The Project Gutenberg eBook of The Rural Magazine, and Literary Evening Fire-Side, Vol. 1 No. 07 (1820)

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: The Rural Magazine, and Literary Evening Fire-Side, Vol. 1 No. 07
(1820)

Author: Various

Release date: April 22, 2015 [eBook #48766]

Most recently updated: October 24, 2024

Language: English

Other information and formats: www.gutenberg.org/ebooks/48766

Credits: Produced by Barbara Tozier, Bill Tozier and the Online
Distributed Proofreading Team at <http://www.pgdp.net>

*** START OF THE PROJECT GUTENBERG EBOOK THE RURAL
MAGAZINE, AND LITERARY EVENING FIRE-SIDE, VOL. 1 NO. 07
(1820) ***

THE
RURAL MAGAZINE,
AND
LITERARY EVENING FIRE-
SIDE.

VOL. I. PHILADELPHIA, *Seventh Month*, 1820. No. 7.

FOR THE RURAL MAGAZINE.

THE VILLAGE TEACHER.

Among the smooth faced urchins that were subject to my little kingdom about fifteen years ago, was a tall awkward boy, named Jonathan Gull. Jonathan was the son of an honest hard working farmer, who lived about two miles from the village, and who had by dint of frugality, acquired some property, and with it a proportional degree of consideration in the eyes of his neighbours. His crops of wheat were generally large, and he made a journey to the metropolis once a year, to dispose of his grain and produce. On these occasions his wife and a grown up daughter would usually accompany him to see the city and to buy cheap goods. It did one's heart good to witness the return of the honest farmer—the smile of self-complacency with which he greeted the members of his family, and the eagerness with which he inquired respecting the farm, Old Roan, and the young colt, and brindle, the cow, and the litter of young pigs; and the air of importance which he assumed towards his neighbours, who thronged around him to hear the latest news—what Boney was doing—the yellow fever—and the price of wheat. His hearty greeting of his acquaintance; the animation which sparkled in his sunburnt face; his short thick set figure, decked out in a suit of homespun grey, with large brass buttons; his arms a kimbo; and the broad burst of merriment, that, amidst the discussion of graver subjects, occasionally broke forth at some sly turn, or second hand joke of the traveller; altogether formed the beau ideal of homely rustic happiness, and prosperity. Nor was the greeting and excitation less on the part of the wife and daughter. As the wagon was emptied of its load, treasure after treasure met the eyes of the delighted group of children and neighbours. Here were a new set of milk pans, and a churn for the dairy; a dozen of pewter spoons, as smooth and as bright as silver, and scrubbing brushes, and knives and forks to repair the waste of years. There glittered lots of new calico, as fine as red and yellow could make them; papers of pins and needles, and all the sundry articles which complete the stock of an industrious housewife; while in another place were cautiously hid, lest they should excite undue envy, the silver teaspoons and teapot, and the bundles of coffee and tea and white sugar, together with the tortoise shell combs and

gold ear-rings which the good natured husband had been importuned to buy. Let not the reader turn away with contempt from this simple picture; the event was an important one in the family of farmer Gull, and supplied it with a stock, not only of necessities and luxuries, but of conversation and pleasure for a full season. But alas! in the train of all this prosperity and gladness marched the forerunners of decay. Farmer Gull's heavy purse of shining dollars had won the heart of many a knight of the counter, and many were the plans laid to obtain a closer intimacy with their owner. Here Mrs. Gull was invited to sit down in the parlour to rest herself; and there was she pressed to stay to tea. One talked to her about her butter and her cheese, and another about her rosy faced children. In short, it so happened, that in the course of a few years, she had at least half a dozen acquaintances in the shopkeeping line, each of whom were under engagement to spend a short time during the Dog-days at Melrose; for so was the farm now styled. With these new acquaintances, new views and expectations filled the minds of the wife and daughters. The old family Bible and the Pilgrim's Progress were less frequently read and quoted; the calico short gowns in which they used to visit their neighbours of a summer evening gave place to dresses of silk and white muslin, and their talk was now of the fashions and the new novels. Among other changes it was determined to make a merchant of Jonathan, and a place was accordingly engaged for him with the particular friend of the family, Mr. Seersucker. —I was struck with the appearance of the lad when he came to bid me good bye and take his books from school. He was dressed in a suit of tidy homespun, made up by the hands of his mother and sisters, and which was, from the sheep that furnished the wool to the thread with which the garments were sewed, the produce of his father's farm. The good old fashioned notions of domestic industry were not then extinct. Our farmers had not then found that it was cheaper to keep a family of strapping daughters idle at home, and send the wool to one factory to be carded and spun, and to another to be wove and dyed, and then to a fop of a tailor to be made up into coats, than to have the cards, the wheel, and the loom at their own fire-side; giving wholesome occupation to their family, enlivening the gloomy hours of winter, and cementing by good humour and mutual assistance the ties of family and kindred. We had not then found out the grand modern secret of economy; that it is better to pamper a few inordinate manufacturing establishments, with all their consequences, of a degraded

and dependant population and high taxes, than to let every individual pursue his own interests, and to encourage that best of domestic manufactures where the workshop is the kitchen of the farmer, and his wife and children the contented and uncorrupted labourers.

But I find myself perpetually digressing from my story, and must bid a truce to these rambling thoughts. Jonathan, as I was saying, was tidily dressed; his hair was combed smooth on his forehead, and hanging long behind, and the awkwardness of his figure, was scarcely apparent in the expression of good health and contentment that animated him. It soon became apparent, however, that citizen Jonathan would not long be contented with his homely garb. At his first visit home I observed that he had been in the hands of a fashionable hair-dresser, who had given him the true Bonapartean topknot. His shirt collar was stretched up till it half covered his ears and cheeks, and he sported a clouded and twisted cane, while the remainder of his dress was yet unchanged. By degrees the exuvia of the clown fell off, as I have seen a snake in the spring slowly emerge from the shrivelled skin which she has cast, or as a locust may sometimes be seen breaking through his faded coat of mail and sporting in gaudy robes of green and gold; or as you may observe in the ponds a kind of doubtful animal, half frog and half tadpole. Jonathan became in a year or two the admiration of all the neighbouring milk maids, and was universally accounted a fine gentleman. He had not been sure ciphered further than Practice, and was but a dull hand, when a boy, at Murray's grammar; but so wonderfully had a city life sharpened his wits, that his old master himself was quite in the back ground when Jonathan was present; he poured forth such a torrent of words and discoursed so fluently about politics and trade and great men. Then came the days of delusion and speculation. Jonathan was now of age, and as he had what was called a good turn for business, and a strong back to support him, he was solicited to engage in trade. A partnership was accordingly formed, and "GULL, SNIPE, & Co." glittered in golden capitals across Market Street. A capital of some thousands was paid down, while goods to the amount of hundreds of thousands were bought and sold.

In the meantime farmer Gull was floating on the very spring tide of prosperity. His wealth, which was yearly increasing, gave him great weight in the neighbourhood; he was made overseer of the poor, and there was a

talk of sending him to Congress. When that ill starred measure which created at a birth a swarm of banks, more greedy and more lean than Pharaoh's kine, was adopted, farmer Gull partook of the delusion. He was made a director of the bank of Potosi, which was located in our village, and from that moment gave himself up to dreams of imaginary wealth. He stocked his farm with merino sheep, at an average of fifty dollars a head, and calculated that in six years he should double his money. Six years have elapsed, and farmer Gull's whole flock will not now sell for the cost of a single ewe. He mortgaged his farm to the bank, that he might buy a neighbouring property, and prosecute some expensive improvements in the way of mills and factories. At home every thing was changed. Mistress Gull rode to church in a handsome carriage, which Jonathan had sent up from town; and Polly and Biddy, instead of being at the milk pail by sunrise, lay abed till breakfast time and then came down with pale and languid countenances, and their hair buckled into "kill-beaus" and "heart breakers," to partake of coffee which unnerved their system, and rendered them feverish and nervous till dinner time. Why should I proceed with my story? The sequel may be read in the present circumstances of many a once thriving family. The lean kine of Simon devoured his fat kine, and distress and confusion covered the face of the country. Gull, Snipe, & Co., after a few years of fictitious prosperity, and proportional extravagance, went the way of half Market Street. Farmer Gull was their security, and had to make heavy payments to the bank. His own speculations had proved ruinous, the clouds became continually darker and thicker around him, and have at length burst upon his head. His whole property is insufficient to pay the mortgages, and his stock and furniture will be sold next week by the sheriff. Such is the termination of farmer Gull's career. His family is incapacitated for its present destitute situation, and has lost the inclination and the power of being frugal: he himself, I observe, bears his troubles with an appearance of unusual fortitude. He preserves his cheerful spirits, and has become the life of a circle of embarrassed farmers that frequent a tavern opposite the window of my study. I see him there daily—sometimes to be sure moody and disconsolate, but more often leading the chorus of some Bacchanalian song, or retailing the merry jests of some quondam acquaintance, of a lawyer, or bank director. Alas! for my countrymen;—when shall we see again the days of honest dealing, sturdy frugality, unsophisticated manners, and household industry?

FOR THE RURAL MAGAZINE.

THE DESULTORY REMARKER.

No. VI.

I've felt full many a heart-ach in my day,
At the mere rustling of a muslin gown,
And caught some dreadful colds, I blush to say,
While shivering in the shade of beauty's frown.
They say her smiles are sunbeams—it may be—
But never a sunbeam would she throw on me.

It has been said by a writer, whose genius and scholarship are in the highest degree honourable to his country, that our Parnassus is fruitful only in weeds, or at best in underwood. Notwithstanding the general correctness of this assertion, a modest wild-flower now and then delights the eye, and points that rainbow adventurer HOPE to the brilliant future; in which some master of song shall disclose in a broad and clear light,

Such forms as glitter in the Muse's ray.

American literature, it must be admitted, is comparatively feeble in many of its branches; but while the names of FRANKLIN and RUSH, DENNIE and BROWN, of WALSH and IRVING, are remembered, it is entitled to the respectful consideration, even of foreign criticism.

The extract given above is made from a writer, who has furnished some evidence of poetic talent in sundry occasional playful pieces, published originally in the New York Evening Post, under the signature of CROAKER & Co. However unprepossessing may be the name which he has chosen to assume, his notes instead of reminding us, as might be inferred, of the frog or the raven; at times successfully rival those of the favourite songsters of the grove. From this stanza, we learn that the author is a BACHELOR; who like too many of his brethren, delights to dwell on the fancied cruelty of the fair; and to pour the unheeded complaints of his sorrows on the dull and listless ear of indifference. To this portion of society, little commiseration

is extended from any quarter; and the general sentiment is responsive to that contained in the following couplet of POPE:

*Let sinful bachelors their woes deplore;
Full well they merit all they feel and more.*

Unpopular as such a doctrine might appear, this condition of life has had its advocates and defenders. Amongst these may be placed that truly great man, the eloquent and accomplished apostle PAUL. When adverting to this subject, he discovers something of an unsocial disposition where he says, "For I would that all men were even as myself."^[1] "But I speak this by permission, and not of commandment," says he, as if fearful that speculative opinion might be received from the weight of his character as authority perfectly valid and conclusive. DRYDEN has asserted, that "a true painter naturally delights in the liberty which belongs to the bachelor's estate." However specious this position may appear, the liberal arts are more indebted to the charities of life, and to the influence of female excellence, than the author of such a sentiment would be willing to admit. A vivid perception of physical and moral beauty, delicacy of feeling, and intellectual refinement, are indispensably requisite in the artist who aspires to eminence in his profession. Nothing has a more direct and efficient tendency to promote elegance and correctness of taste than the society of enlightened and polished females. The absence of care is another immunity which the BACHELOR is said to enjoy; but this as well as other assumptions in his favour, but serves to illustrate the fact, that on almost any subject whatever, to use the language of Sir ROGER DE COVERLY, much may be said on both sides."

In all ages there have been from various causes, a formidable array of individuals of this class. The circumstances connected with the present times are, unfortunately, well calculated to increase their number. To the usual disastrous consequences produced by "beauty's frown," the disappointments and gloomy prospects in business, deep rooted habits of idleness and extravagance, by which the present period is peculiarly distinguished, may also be added. Active industry, frugality, and temperance, should be sedulously cultivated as moral virtues; having a most important agency in augmenting the stock of individual, social, and political

happiness. But unamiable and repulsive as the character of a BACHELOR may too frequently be, is it necessarily so?

Can he contemplate the condition of childhood, surrounded with the pallid spectres of poverty, and shooting forth luxuriantly into all the noxious forms of ignorance and vice;—can he walk our streets or wharves when his ear is saluted with their lisping imprecations;—or witness their utter disregard of the duties which appertain to the Sabbath,—without seriously interrogating his own bosom—In what way can any exertions of mine improve their condition, and promote their true interests?

Can he behold the increase of intemperance and crime in all the ramifications of society, without feeling the influence of those sacred ties which bind him to that community of which he is a member; and without resolving to use all diligence to arrest their further extension, so far as his influence and example may reach?

Can he listen with unconcern to the cries of oppressed humanity, and view without emotion, those objects of wretchedness which almost daily present themselves in the most affecting shapes, and forget the intimate relationship, and the reciprocity of duties which exist between every branch of the human family, and the justice and force of the claims of distress upon every generous and sympathetic heart?

He can nowhere in the moral or physical economy of the world, find an example of existences which are independent of all connection with the present, past, and future. The universe has been with great propriety compared to a complex machine, "a stupendous whole," every part of which has its relative and proper function to perform, and discord and confusion are the consequence of each irregularity of movement.

From Nature's chain whatever link you strike,
Tenth, or ten thousandth breaks the chain alike.

It should be the business of *every one* to cultivate such sentiments as those which are contained in the extract below, given from a work^[2] which the celebrated DUGALD STEWART declared when presenting a copy of it to one of our countrymen, now a resident of Philadelphia, to be the finest piece of composition in the English language: "*That to feel much for others,*

and little for ourselves, that to restrain our selfish, and to indulge our benevolent affections, constitutes the perfection of human nature; and can alone produce among mankind that harmony of sentiments and passions in which consists their whole grace and propriety." Even a BACHELOR, actuated by principles of this character, might emerge into the character of a worthy, useful, and amiable man!



Extracts from C. E.'s Common-Place Book.

Cheap food for horses, from M'Arthur's Financial Facts, 8vo. p. 258.

The author lived in London and kept three horses which he fed as follows.

Two trusses and a half of clover or meadow hay cut and mixed with four trusses of wheat, or barley straw, when cut up, make nearly equal quantities in weight; two heaped bushels of this mixture equal to fourteen pounds weight, are given to each horse in twenty-four hours, being previously mixed with half a peck of corn, (in England oats) ground or chopped, weight 5lbs. with water to wet it; that is, 7 pounds of hay; 7 pounds of straw; 5 pounds of meal; given at six different times, each day and evening. Add 5 pounds of hay at night, makes 24 pounds to each horse in twenty-four hours; and it kept them much fatter than with double the corn each day unground, two trusses of hay a week.

An ox, unworked, eats about 32 pounds of meadow hay per day.

An ox at work eats 40 pounds a day.

If fed in the stable each head of horned cattle will eat 130 pounds green clover just cut, or 30 pounds clover hay a day.

At work, 3 horses eat in all, 48 stones a week of hay, also 48 quarts of oats a week each horse.

At work 18 horses in 12 days, eat 430 stones of hay, which is 14 stones a week for each horse, also 64 quarts of oats a week for each.

An idle horse eat 14 stones of hay a week and no corn.

Native Grape Vine.

AUGUST 22, 1807.

In the garden of Joseph Cooper, Esq. of New Jersey, just opposite to Philadelphia, is one grape vine which with its branches, covers 2170 square feet of ground. On this one vine are now grapes supposed to be forty bushels, and probably much more. It produced last year one barrel of wine, which was made without sugar, and is judged to be quite as good as Madeira of the same age, by a man brought up in the Madeira wine trade. Under this vine the ground produced a good crop of grass this season. It is a native American vine, transplanted from that same neighbourhood. ^[3]

If 2170 square feet produced 32 gallons, then one acre which is 43,560 square feet would produce about 20 barrels, or 640 gallons but allowing space for avenues, say about 15 barrels, or 480 gallons.

It is expected that the crop of grapes for 1807 will produce much more than those of 1806.

One acre yielding 480 gallons,
at \$1.00, is - - - \$480.00
at \$1.50, is - - - \$720.00

This holds out a profitable culture to farmers.

C. E.

TO THE EDITORS OF THE RURAL MAGAZINE.

The "*Philadelphia Society for Promoting Agriculture*," at a late meeting, passed a resolution to recommend the use of Malt Liquors, in preference to Ardent Spirits, on farms;—and appointed the subscribers a Committee to procure and publish Directions to enable Farmers to Brew Beer.—They have accordingly the pleasure to send a pamphlet published by the proprietors of a patent English brewing apparatus, which was imported by a gentleman of Philadelphia; and also some directions by an eminent brewer, to enable families to brew beer with the common household utensils. The apparatus was tried last year by one of our members, and found to answer perfectly.—It was imported with the view to general utility, not to private profit, and we understand may be purchased at first cost.^[4]

RICHARD PETERS, }
JAMES MEASE, }
ROBERTS VAUX, }
ISAAC C. JONES, } *Committee.*

Philadelphia, June, 1820.

DIRECTIONS FOR BREWING

With Needham & Co.'s Patent Portable Family Brewing Machine.

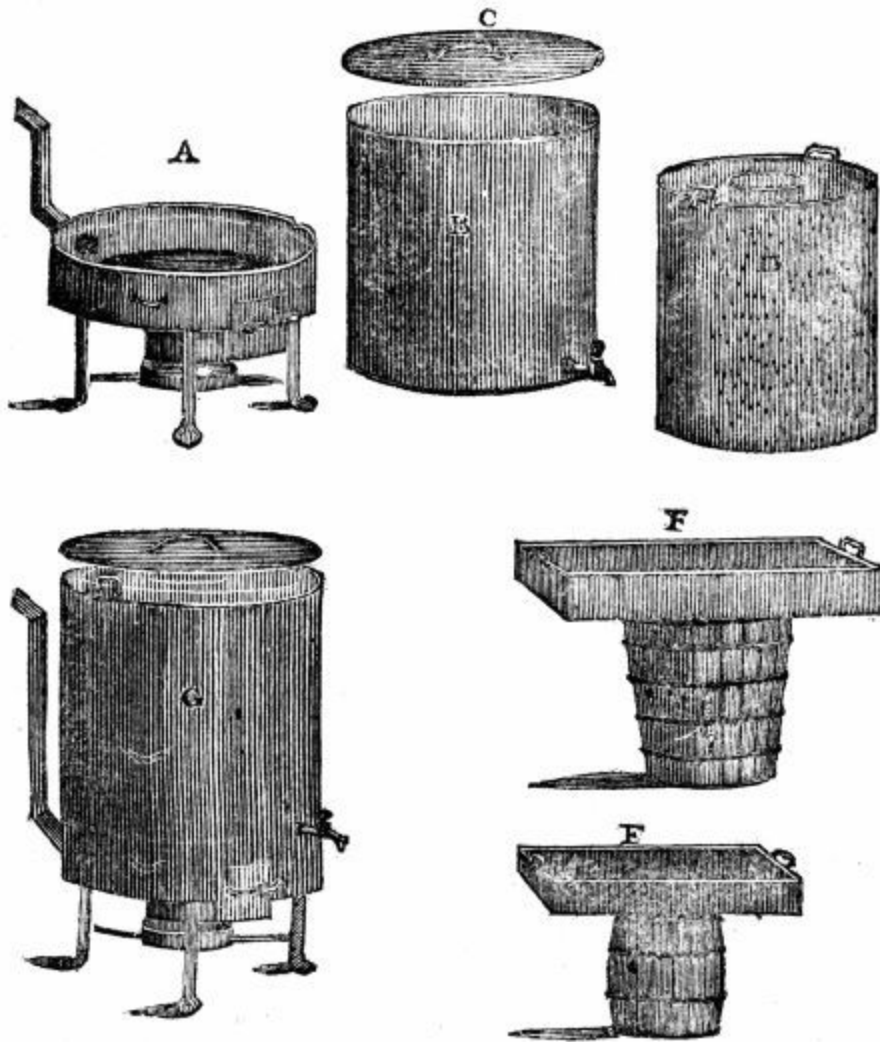
As the attainment of good Malt Liquor greatly depends upon the quality of the materials from which it is produced, it may be useful to give a few general instructions for distinguishing the quality of malt and hops, of which it should be only composed; but considerable practice being requisite to form a ready judgment, it will generally be more safe to buy them of some reputable dealer.

MALT.—To judge of the quality of malt, you must chew it, and if *sweet, tender, and mealy, with a brisk full flavour*, it is good; in coloured malt particular care should be taken that it is neither smoky nor burnt. ^[5]

HOPS should be of a *bright colour, free from green leaves, of a quick pungent smell, and glutinous quality*, which will be discoverable by their adhering together, and by rubbing them in the hands. *New hops are preferable to old, after Christmas.*

To Brew Ale with Table Beer after, from the same malt and hops.—The malt should be pale, sweet, and tender, ground coarse, and the hops of a pale bright colour and glutinous quality.

If the ale is for present use, $\frac{3}{4}$ of a pound of hops to each bushel of malt will be sufficient, but for store ale use one pound per bushel.



The machine being placed ready for use as described in the plate, figure G,^[6] put into it as much cold water as will cover the perforated bottom of the extracting cylinder, and light the fire; then put as much coarse ground malt into the perforated cylinder, (see the plate, figure D,) as will three parts fill it, taking care that none goes into the centre, (which centre should be covered, *but only*) while putting the malt in, and when mashing the malt,) nor any between the cylinder and boiler.—The malt being put in, pour through the centre as much more cold water as will just cover the malt, then make the fire good, and in one hour after stir the malt well up with a strong mashing stick, for about ten minutes, so that every particle of malt may be divided from the other: let the heat increase to 180 degrees, which you must ascertain by holding the thermometer a minute in the centre part of the machine, and when at 180 degrees of heat, stir the malt again, and after this

second stirring, try the heat, and if then at 180 degrees, damp the fire well with some wet ashes to prevent the heat of the mash from increasing, and in 3 hours and a half from the time of lighting the fire, draw off the wort very gently that it may run fine, and put it into one of the coolers, and put all the hops (rubbing them through your hands to break the lumps) on the top of the wort to keep it hot till the time for returning it into the machine for boiling; having drawn off this ale wort, put into the machine through the centre as much more cold water as will cover the grains, brisken the fire, and in half an hour stir up the malt for about ten minutes, and make it 180 degrees of heat as quick as you can, then damp the fire to prevent its getting hotter, and in one hour and a half from the time of putting in the water, draw off this table beer wort gently, that it may run fine, and put it into the other cooler, and cover it over to keep it hot until the time for returning it into the machine for boiling; having drawn off this table beer wort, if you wish to make a third wort, put in as much more cold water into the machine as you think proper, and make it 170 degrees of heat as quick as you can, and draw it off in about an hour after, and put it to the last drawing off, or wort: then take the grains out of the cylinder with a hand shovel as clean as you can, and after, take out the cylinder,^[2] and with a birch broom and a little water rince out the boiler clean, and put back the perforated cylinder into the boiler, and then put the first drawing off or ale wort, with all the hops, into the machine cylinder where you have taken the grains from, and cover the machine, but be sure the centre cover is off; make it boil as quick as you can, and let it boil well one hour, then damp the fire and draw it off into a cooler or coolers, which should be placed in the air where it will cool quick. Having drawn off this ale wort, return the second drawing off, or table beer wort, with the third, into the machine to the hops left from the ale wort, stir up the fire and make it boil as quick as you can, and let it boil well one hour, then put out the fire and draw off the wort, and put it into a cooler placed in the air to cool quick; when the worts in the coolers are cooled down to 70 degrees of heat by the thermometer, put the proportion of a gill of fresh thick yeast to every 9 gallons of wort in the coolers, first thinning the yeast with a little of the wort before you put it in that it may the better mix; and when the ale wort is cooled down to 60 degrees of heat, draw it off from the coolers with the yeast and sediment, and put it into the machine boiler (the machine boiler having been previously cleared from the hops and cylinder,) which forms a convenient vessel placed on its stand for the

ale to ferment in, which must be kept fermenting in it with the cover off until the head has the appearance of a thick brown yeast on the surface, an inch or two deep, which will take 3 or 4 days;^[8] when the head has this appearance, draw off the beer free from the yeast and bottoms into a clean cask, which must be filled full,^[9] and when done working, put in a handful of dry hops, bung it down tight, and stow it in a cool cellar. This ale will be fit to tap in 3 or 4 weeks.

The second wort for table beer should be put from the coolers with yeast and sediment into an upright cask, with the cover off or top head out, at not exceeding 60 degrees of heat, and as soon as you perceive a brown yeast on the surface, draw it off free from the yeast and bottoms into a clean cask, which must be kept filled full, and when done working, put in a handful of dry hops, bung it down tight, and stow it in a cool cellar. This table beer will be fit to tap in a week, or as soon as fine.

To make Table Ale.—Mix the first and second worts together, and ferment, and treat it the same as the ale.

To Brew Porter or Brown Beer, with Table Beer after, from the same Malt and Hops.—Use pale and brown malt in equal quantities, ground coarse; and strong brown coloured hops of a glutinous quality. If the beer is for present draught, $\frac{3}{4}$ pound of hops to each bushel of malt will be sufficient, but if intended for store beer, use one pound to each bushel of malt.

The process of brewing is the same as described for brewing ale with table beer after, except the heat of each mash must not be so high by 10 degrees, on account of the brown malt; the first wort fermented by itself will be stout porter, and fit to tap in 3 or 4 weeks; the second wort will be the table beer, and fit to tap in a week, or as soon as fine; but if you mix the first and second worts together, the same as for table ale, it will be good common porter.

To Brew Table Beer only.—Let your malt be of one sort, of a full yellow colour (not brown malt) ground coarse, and strong brown coloured hops, of a glutinous quality. If for present draught $\frac{1}{2}$ a pound of hops to each bushel of malt will be sufficient, but if for keeping two or 3 months, use one pound of hops per bushel.

The process of brewing is the same as described for brewing porter and table beer, with the addition of another wort, that is, filling the machine a third time with water before you take out the grains, and treating the third mash the same as the second.

The first drawing off or wort, with part of the second wort, to be boiled (first) one hour with all the hops, and the remainder of the second wort with the third, to be boiled next one hour to the same hops; these two boilings, when cooled down to 60 degrees of heat, (having put your yeast to it in the coolers at 70 degrees) must be put together to ferment in the machine boiler, and as soon as it has the appearance of a brown yeast on the surface, draw it off into the casks, which must be kept filled full, and when done working, put into each cask a handful of dry hops, bung it down tight, and put it into a cool cellar. Tap it in a week, or as soon as fine.

General Remarks.—The season for brewing sound keeping beer, is from October to May.

All Beer should be stored in cool cellars or vaults, and kept as much as possible from the common atmosphere; and in drawing beer from a cask, if necessary to raise the vent-peg, it should be carefully tightened as soon as the beer is drawn.

When beer is intended to be kept many months, the bungs of the casks, and if bell casks are used, the whole of the head should be covered with sand or clay, which should be kept moist.

To preserve the Machine.—When the brewing is over, wash the machine and coolers with cold or hot water, then dry them, and put them away in a dry place. When wanted to be used, they should be washed with boiling water.

To keep Casks sweet.—It is recommended when a cask of beer is drawn off, to take out the head and scrub out the cask; then thoroughly dry and put it away in a dry place with the head out.

If it should be inconvenient to take out the head, and the cask is wanted to be filled again quickly, it may be washed quite clean with warm water, and afterwards with lime water; or the grounds being left in the cask, and every vent stopped, (bung, tap, and vent holes,) it may be kept in that state for a short time.

Casks of a bell shape are preferable for private brewing, and the patentees make them upon a principle by which the inside can be scrubbed out clean without removing the head, and at the same price as common casks of that shape.

The following comparative statement of the cost of Brewing Beer with Needham's apparatus, and of Beer when purchased, is given by the proprietor.—The references are to London prices, but an opinion may still be formed of the great economy of Domestic Brewing.

Daily Consumption of	Yearly Consumption	Brewers' Prices.	Yearly expense
ale in a family, 3 pints	is 137 gallons	At 2s. 6d. per gal.	L.17 2 6
Do. of table beer, 3 pints	Is 137 gallons	At 8d.	4 11 4
			21 13 10

To Brew the above quantity of good Ale and Table Beer, $15\frac{1}{4}$ Bushels of Malt are required, the L.7 12 6 cost of which, at 10s. per Bushel, will be

11½ lbs. best Hops, for ditto, at 5s.	2176	10 0
Yearly saving in the above quantity		11 3 10

The above calculation sufficiently proves that the Patent Brewing Machine will, to the smallest Family who purchase Brewer's Beer, pay for itself in One Year, and those who have been accustomed to Brew by the old Method, will find the beer much stronger and better by using this Machine, and very considerably less likely to be spoiled in Brewing, with a great saving in Fuel, Labour, and Time.

As it may be inconvenient, or too expensive, for many private families to purchase a brewing machine, the following Directions are subjoined, which will enable them, by the aid of the vessels used in a family, to brew a barrel of beer; and by attention, and a few experiments, they will produce an excellent beverage.

Prepare a tub for making the extract, by fitting a false bottom with numerous holes, and raised about half an inch from the real bottom, in which fix a cock for drawing off the extract. Have four bushels of malt coarsely ground, and heat your water to about 170 or 175 degrees^[10] of heat, of Fahrenheit's thermometer. Then pour in the tub about thirty-eight gallons of the water, and gently stir in the malt, until it is all mixed. Cover it, and let it stand about an hour and a half; draw off the extract into a vessel, and throw in about one and a quarter pounds of hops for liquor for present use, or about two or two and a half pounds for keeping liquors; cover the vessel to keep in the heat, and pour over the malt about 26 gallons of water, of about the same heat as the first, stirring it until it is well mixed with the malt; let it stand one hour, then draw off the extract, add it to the first extract, and put them on to boil in an open kettle: this will be your strong beer. Then pour over the malt about twenty gallons of water, for small beer, at about 160 or 170 degrees of heat. This last will not require stirring, and the extract may remain covered until the kettle is ready for it. Keep the strong beer boiling smartly for about one hour and a quarter, or one hour and a half, for present use, or two hours for keeping: then pour it through a sieve or strainer, and set it to cool. Return the hops into the kettle with your third extract, or small beer, which set to boiling as soon as practicable, and continue it for about an hour and a half; then pour it through the strainer, and set it to cool. When cool, ferment according to the directions accompanying the brewing machine. The quantity of water used may be varied at the discretion of the person brewing. By diminishing the water, he may increase the strength of the liquor, or by increasing it, diminish the strength. Thus with the hops he may vary the quantity to suit his palate in the degree of bitter flavour that may be most agreeable.—For fomenting, a cask with one head taken out, will answer the same purpose as the machine boiler.

THE SNOW STORM.

From Blackwood's Edinburgh Magazine, for April.

"'Tis only from the belief of the goodness and wisdom of a Supreme Being, that our calamities can be borne in that manner which becomes a man."—HENRY MACKENZIE.

In summer there is a beauty in the wildest moors of Scotland, and the wayfaring man who sits down for an hour's rest beside some little spring that flows unheard through the brightened moss and water-cresses, feels his weary heart revived by the silent, serene, and solitary prospect. On every side sweet sunny spots of verdure smile towards him from among the melancholy heather—unexpectedly in the solitude a stray sheep, it may be with its lambs, starts half alarmed at his motionless figure—insects large, bright, and beautiful come careering by him through the desert air—nor does the Wild want its own songsters, the grey linnet, fond of the blooming furze, and now and then the lark mounting up to heaven above the summits of the green pastoral hills.—During such a sunshiny hour, the lonely cottage on the waste seems to stand a paradise; and as he rises to pursue his journey, the traveller looks back and blesses it with a mingled emotion of delight and envy. There, thinks he, abide the children of innocence and contentment, the two most benign spirits that watch over human life.

But other thoughts arise in the mind of him who may chance to journey through the same scene in the desolation of Winter. The cold bleak sky girdles the moor as with a belt of ice—life is frozen in air and on earth. The silent is not of repose but extinction—and should a solitary human dwelling catch his eye half buried in the snow, he is sad for the sake of them whose destiny it is to abide far from the cheerful haunts of men, shrouded up in melancholy, by poverty held in thrall, or pining away in unvisited or untended sickness.

But, in truth, the heart of human life is but imperfectly discovered from its countenance; and before we can know what the summer, or what the winter yields for enjoyment or trial to our country's peasantry, we must have conversed with them in their fields and by their fire-sides; and make ourselves acquainted with the powerful ministry of the seasons, not over

those objects alone that feed the eye and the imagination, but over all the incidents, occupations, and events that modify or constitute the existence of the poor.

I have a short and simple story to tell of the winter life of the moorland cottager—a story but of one evening—with a few events and no signal catastrophe—but which may haply please those hearts whose delight it is to think on the humble under-plots that are carrying on in the great Drama of life.

Two cottagers, husband and wife, were sitting by their cheerful peat-fire one winter evening, in a small lonely hut, on the edge of a wide moor, at some miles distance from any other habitation. There had been, at one time, several huts of the same kind erected close together, and inhabited by families of the poorest class of day-labourers who found work among the distant farms, and at night returned to dwellings which were rent free, with their little gardens, won from the waste.—But one family after another had dwindled away, and the turf-built huts had all fallen into ruins, except one that had always stood in the centre of this little solitary village, with its summer walls covered with the richest honeysuckles and in the midst of the brightest of all the gardens. It alone now sent up its smoke into the clear winter sky—and its little end window, now lighted up, was the only ground star that shone towards the belated traveller, if any such ventured to cross, on a winter night, a scene so dreary and desolate. The affairs of the small household were all arranged for the night. The little rough pony that had drawn in a sledge, from the heart of the Black-Moss, the fuel by whose blaze the cottiers were now sitting cheerily and the little Highland cow, whose milk enabled them to live, were standing amicably together, under cover of a rude shed, of which one side was formed by the peat stack, and which was at once byre, and stable, and hen-roost. Within, the clock ticked cheerfully as the fire-light readied its old oak-wood case across the yellow sanded floor—and a small round table stood between, covered with a snow-white cloth, on which were milk and oat cakes, the morning, mid-day, and evening meal of these frugal and contented cottiers. The spades and the mattocks of the labourer were collected into one corner, and showed that the succeeding day was the blessed Sabbath—while on the wooden chimney-piece were seen lying an open Bible ready for family worship.

The father and mother were sitting together without opening their lips, but with their hearts overflowing with happiness, for on this Saturday-night they were, every minute, expecting to hear at the latch the hand of their only daughter, a maiden of about fifteen years, who was at service with a farmer over the hills. This dutiful child was, as they knew, to bring home to them "her sair-worn penny fee," a pittance which, in the beauty of her girlhood, she earned singing at her work, and which, in the benignity of that sinless time, she would pour with tears into the bosoms she so dearly loved. Forty shillings a year were all the wages of sweet Hannah Lee—but though she wore at her labour a tortoise-shell comb in her auburn hair, and though in the kirk none were more becomingly arrayed than she, one half, at least, of her earnings were to be reserved for the holiest of all purposes, and her kind innocent heart was gladdened when she looked on the little purse that was, on the long expected Saturday-night, to be taken from her bosom, and put, with a blessing, into the hand of her father, now growing old at his daily toils.

Of such a child the happy cottiers were thinking in their silence. And well might they be called happy. It is at that sweet season that filial piety is most beautiful.—Their own Hannah, had just outgrown the mere unthinking gladness of childhood, but had not yet reached that time, when inevitable selfishness mixes with the pure current of love. She had begun to think on what her affectionate heart had felt so long; and when she looked on the pale face and bending frame of her mother, on the deepening wrinkles, and whitening hairs of her father, often would she lie weeping for their sakes on her midnight bed—and wish that she was beside them as they slept, that she might kneel down and kiss them, and mention their names over and over again in her prayer. The parents whom before she had only loved, her expanding heart now also venerated. With gushing tenderness was now mingled a holy fear and an awful reverence. She had discerned the relation in which she, an only child, stood to her poor parents now that they were getting old, and there was not a passage in Scripture that spake of parents or of children, from Joseph sold into slavery, to Mary weeping below the Cross, that was not written, never to be obliterated, on her uncorrupted heart.

The father rose from his seat, and went to the door to look out into the night.—The stars were in thousands—and the full moon was risen.—It was

almost light as day, and the snow, that seemed incrustated with diamonds, was so hardened by the frost, that his daughter's homeward feet would leave no mark on its surface. He had been toiling all day among the distant Castle-woods, and, stiff and wearied as he now was, he was almost tempted to go to meet his child, but his wife's kind voice dissuaded him, and returning to the fire-side, they began to talk of her whose image had been so long passing before them in their silence.

"She is growing up to be a bony lassie," said the mother, "her long and weary attendance on me during my fever last spring kept her down awhile—but now she is sprouting fast and fair as a lily, and may the blessing of God be as dew and as sunshine to our sweet flower all the days she bloometh upon this earth." "Aye Agnes," replied the father, "we are not very old yet—though we are getting older—and a few years will bring her to women's estate, and what thing on this earth, think ye, human or brute, would ever think of injuring her? Why I was speaking about her yesterday to the minister as he was riding by, and he told me that none answered at the Examination in the Kirk so well as Hannah.—Poor thing—I well think she has all the Bible by heart—indeed, she has read but little else—only some stories, too true ones, of the blessed martyrs, and some o' the auld sangs o' Scotland, in which there is nothing but what is good, and which, to be sure, she sings, God bless her, sweeter than any laverock." "Aye—were we both to die this very night, she would be happy—not that she would forget us, all the days of her life. But have you not seen, husband, that God always makes the orphan happy?—None so little lonesome as they! They come to make friends o' all the bonny and sweet things in the world around them, and all the kind hearts in the world make friends o' them. They come to know that God is more especially the father o' them on earth whose parents he has taken up to heaven—and therefore it is that they for whom so many have fears, fear not at all for themselves, but go dancing and singing along like children whose parents are both alive! Would it not be so with our dear Hannah? So douce and thoughtful a child—but never sad nor miserable—ready it is true to shed tears for little, but as ready to dry them up and break out into smiles! I know not why it is, husband, but this night my heart warms toward her, beyond usual. The moon and stars are at this moment looking down upon her, and she looking up to them, as she is glinting homewards over the snow. I wish she were but here, and taking the comb

out o' her bonny hair, and letting it all fall down in clusters before the fire, to melt away the cranreuch!"

While the parents were thus speaking of their daughter a loud sugh of wind came suddenly over the cottage, and the leafless ash-tree under whose shelter it stood, creaked and groaned dismally, as it passed by.—The father started up, and going again to the door, saw that a sudden change had come over the face of the night. The moon had nearly disappeared, and was just visible in a dim yellow, glimmering in the sky.—All the remote stars were obscured, and only one or two faintly seemed in a sky that half an hour before was perfectly cloudless, but that was now driven with rack, and mist, and sleet, the whole atmosphere being in commotion. He stood for a single moment to observe the direction of this unforeseen storm, and then hastily asked for his staff. "I thought I had been more weather-wise—A storm is coming down from the Cairnbrae-hawse, and we shall have nothing but a wild night." He then whistled on his dog—an old sheep dog, too old for its former labours and set off to meet his daughter, who might then, for aught he knew, be crossing the Black-moss.

The mother accompanied her husband to the door, and took a long frightened look at the angry sky. As she kept gazing, it became still more terrible. The last shred of blue was extinguished—the wind went whirling in roaring eddies, and great flakes of snow circled about in the middle air, whether drifted up from the ground, or driven down from the clouds, the fear-stricken mother knew not, but she at least knew, that it seemed a night of danger, despair, and death. "Lord have mercy on us James, what will become of our poor bairn!" But her husband heard not her words, for he was already out of sight in the snow storm, and she was left to the terror of her own soul in that lonesome cottage.

Little Hannah Lee had left her master's house, soon as the rim of the great moon was seen by her eyes, that had been long anxiously watching it from the window, rising like a joyful dream, over the gloomy mountain-tops; and all by herself she tripped along beneath the beauty of the silent heaven. Still as she kept ascending and descending the knolls that lay in the bosom of the glen, she sung to herself a song, a hymn, or a psalm, without the accompaniment of the streams, now all silent in the frost; and ever and anon she stopped to try to count the stars that lay in some more beautiful part of the sky, or gazed, on the constellations that she knew, and called them, in

her joy, by the names they bore among the shepherds.—There were none to hear her voice, or see her smiles, but the ear and eye of Providence. As on she glided and took her looks from heaven, she saw her own little fire-side—her parents waiting for her arrival—the bible opened for worship—her own little room kept so neatly for her, with its mirror hanging by the window, in which to braid her hair by the morning light—her bed prepared for her by her mother's hand—the primroses in her garden peeping through the snow—old Tray, who ever welcomed her home with his dim white eyes—the poney and the cow; friends all, and inmates of that happy household. So stepped she along, while the snow diamonds glittering around her feet, and the frost wove a wreath of lucid pearls around her forehead.

She had now reached the edge of the Black-moss, which lay half way between her master's and her father's dwelling, when she heard a loud noise coming down from Glen-Scrae, and in a few seconds, she felt on her face some flakes of snow. She looked up the glen, and saw the snow storm coming down, fast as a flood. She felt no fears; but she ceased her song; and had there been a human eye to look upon her there, it might have seen a shadow on her face. She continued her course, and felt bolder and bolder every step that brought her nearer to her parent's house. But the snow storm had now reached the Black-moss, and the broad line of light that had lain in the direction of her home, was soon swallowed up, and the child was in utter darkness. She saw nothing but the flakes of snow, interminably intermingled, and furiously wafted in the air, close to her head; she heard nothing but one wild, fierce, fitful howl. The cold became intense, and her little feet and hands were fast, being benumbed into insensibility.

"It is a fearful change," muttered the child to herself, but still she did not fear, for she had been born in a moorland cottage, and lived all her days among the hardships of the hills.—"What will become of the poor sheep," thought she,—but still she scarcely thought of her own danger, for innocence and youth, and joy, are slow to think of ought evil befalling themselves, and thinking benignly of all living things, forget their own fear in their pity of others' sorrow.—At last, she could no longer discern a single mark on the snow, either of human steps, or of sheep track, or the foot print of a wild-fowl. Suddenly, too, she felt out of breath and exhausted,—and shedding tears for herself at last sank down in the snow.

It was now that her heart began to quake for fear. She remembered stories of the shepherds lost in the snow,—of a mother and child frozen to death on that very moor,—and, in a moment she knew that she was to die. Bitterly did the poor child weep, for death was terrible to her, who, though poor, enjoyed the bright little world of youth and innocence. The skies of heaven were dearer than she knew to her,—so were the flowers of earth. She had been happy at her work—happy in her sleep—happy in the kirk on Sabbath. A thousand thoughts had the solitary child,—and in her own heart was a spring of happiness, pure and undisturbed as any fount that sparkles unseen all the year through, in some quiet nook among the pastoral hills. But now there was to be an end of all this,—she was to be frozen to death—and lie there till the thaw might come; and then her father would find her body, and carry it away to be buried in the kirk-yard.

The tears were frozen on her cheeks as soon as shed—and scarcely had her little hands strength to clasp themselves together, as she thought of an overruling and merciful Lord came across her heart. Then, indeed, the fears of this religious child were calmed, and she heard without terror, the plover's wailing cry, and the deep boom of the bittern sounding in the moss. "I will repeat the Lord's prayer." And drawing her plaid more closely around her, she whispered beneath its ineffectual cover; "Our Father which art in heaven, hallowed be thy name,—thy kingdom come—thy will be done on earth as it is in heaven." Had human aid been within fifty yards, it could have been of no avail—eye could not see her—ear could not hear her in that howling darkness. But that low prayer was heard in the centre of eternity—and that little sinless child was lying in the snow, beneath the all-seeing eye of God.

The maiden having prayed to her Father in Heaven—then thought of her Father on earth. Alas! they were not far separated! The father was lying but a short distance from his child; he too had sunk down in the drifting snow, after having in less than an hour, exhausted all the strength of fear, pity, hope, despair, and resignation, that could rise in a father's heart, blindly seeking to rescue his only child from death, thinking that one desperate exertion might enable them to die in each other's arms. There they lay, within a stone's throw of each other, while a huge snow drift was every moment piling itself up into a more insurmountable barrier between the dying parent and his dying child.

There was all this while a blazing fire in the cottage—a white spread table—and beds prepared for the family to lie down in peace. Yet was she who sat thereon more to be pitied than the old man and the child stretched upon the snow. "I will not go to seek them—that would be tempting Providence—and wilfully putting out the lamp of life. No! I will abide here, and pray for their souls?" Then as she knelt down, looked she at the useless fire burning away so cheerfully, when all she loved might be dying of cold—and unable to bear the thought, she shrieked out a prayer, as if she might pierce the sky up to the very throne of God, and send with it her own miserable soul, to plead before Him for the deliverance of her child and husband.—She then fell down in blessed forgetfulness of all trouble, in the midst of the solitary cheerfulness of that bright-burning hearth—and the bible, which she had been trying to read in the pause of her agony, remained clasped in her hands.

Hannah Lee had been a servant for more than six months—and it was not to be thought that she was not beloved in her master's family. Soon after she had left the house, her master's son, a youth of about eighteen years, who had been among the hills, looking after the sheep, came home, and was disappointed to find that he had lost an opportunity of accompanying Hannah part of the way to her father's cottage. But the hour of eight had gone by, and not even the company of young William Grieve, could induce the kind-hearted daughter to delay sitting out on her journey, a few minutes beyond the time promised to her parents. "I do not like the night," said William—"there will be a fresh fall of snow soon, or the witch of Glen Scrae is a liar, for a snow-cloud is hanging o'er the birch-tree-linn, and it may be down to the Black-moss as soon as Hannah Lee." So he called his two sheep dogs, that had taken their place under the long table before the window, and set out, half in joy, half in fear, to overtake Hannah and see her safely across the Black-moss.

The snow began to drift so fast, that before he had reached the head of the Glen, there was nothing to be seen but a little bit of the wooden rail of the bridge across the Sauch burn. William Grieve was the most active shepherd in a large pastoral parish—he had often passed the night among the wintery hills for the sake of a few sheep, and all the snow that had ever fell from Heaven, would not have made him turn back when Hannah Lee was before him; and as his terrified heart told him, in imminent danger of

being lost.—As he advanced, he felt that it was no longer a walk of love or friendship, for which he had been glad of an excuse. Death stared him in the face, and his young soul, now beginning to feel all the passions of youth, was filled with phrenzy. He had seen Hannah every day—at the fire-side—at work—in the kirk—on holydays—at prayers—bringing supper to his aged parents—smiling and singing about the house from morning till night. She had often brought his own meal to him among the hills—and he now found, that though he had never talked to her about love, except smilingly, and playfully, that he loved her beyond father or mother, or his own soul. "I will save thee, Hannah," he cried with a loud sob, "or lie down beside thee in the snow—and we will die together in our youth." A wild whistling wind went by him, and the snow-flakes whirled so fiercely round his head, that he staggered on for a while in utter blindness. He knew the path that Hannah must have taken, and went forward shouting aloud, and stopping every twenty yards to listen for a voice. He sent his well trained dogs over the snow in all directions—repeating to them her name "Hannah Lee," that the dumb animals might, in their sagacity, know for whom they were searching; and as they looked up in his face, and set off to scour the moor, he almost believed that they knew his meaning, (and it is probable they did) and were eager to find in her bewilderment the kind maiden by whose hand they had so often been fed. Often went they off into the darkness, and as often returned, but their looks showed that every quest had been in vain. Meanwhile the snow was of a fearful depth, and falling without intermission or diminution. Had the young shepherd been thus alone, walking across the moor on his ordinary business, it is probable that he might have been alarmed for his own safety—nay, that in spite of all his strength and agility, he might have sunk down beneath the inclemency of the night and perished. But now, the passion of his soul carried him with supernatural strength along, and extricated him from wreath and pitfall. Still there was no trace of poor Hannah Lee—and one of his dogs at last came close to his feet, worn out entirely, and afraid to leave its master—while the other was mute, and, as the shepherd thought, probably unable to force its way out of some hollow or through some floundering drift.—Then he all at once knew that Hannah Lee was dead—and dashed himself down in the snow in a fit of passion.

It was the first time that the youth had ever been sorely tried—all his hidden and unconscious love for the fair lost girl had flowed up from the

bottom of his heart—and at once the sole object which had blessed his life and made him the happiest of the happy, was taken away and cruelly destroyed—so that sullen, wrathful, baffled and despairing, there he lay, cursing his existence, and in too great agony to think of prayer. "God," he then thought, "has forsaken me—and why should he think on me, when he suffers one so good and beautiful as Hannah to be frozen to death." God thought both of him and Hannah—and through his infinite mercy forgave the sinner in his wild turbulence of passion. William Grieve had never gone to bed without joining in prayer—and he revered the Sabbath-day and kept it holy. Much is forgiven to the human heart by him who so fearfully framed it; and God is not slow to pardon the love which one human being bears to another, in his frailty—even though that love forget or arraign his own unsleeping providence. His voice has told us to love one another—and William loved Hannah in simplicity, innocence, and truth. That she should perish, was a thought so dreadful, that, in its agony God seemed a ruthless being—"blow—blow—blow—and drift us up for ever—we cannot be far asunder—O Hannah—Hannah—think ye not that the fearful God has forsaken us?"

As the boy groaned these words passionately through his quivering lips, there was a sudden lowness in the air, and he heard the barking of his absent dog, while the one at his feet hurried off in the direction of the sound, and soon loudly joined the cry. It was not a bark of surprise—or anger—or fear—but of recognition and love. William sprung up from his bed in the snow and with his heart knocking at his bosom even to sickness, he rushed headlong through the drifts, with a giant's strength, and fell down half dead with joy and terror beside the body of Hannah Lee.

But he soon recovered from that fit, and lifting the cold corpse in his arms, he kissed her lips, and her cheeks, and her forehead, and her closed eyes, till, as he kept gazing on her face in utter despair, her head fell back on his shoulder, and a long deep sigh came from her inmost bosom.—"She is yet alive thank God!"—and as that expression left his lips for the first time that night, he felt a pang of remorse:" "I said, O God, that thou hadst forsaken us—I am not worthy to be saved; but let not this maiden perish, for the sake of her parents, who have no other child."

The distracted youth prayed to God with the same earnestness as if he had been beseeching a fellow creature, in whose hand was the power of life

and of death. The presence of the Great Being was felt by him in the dark and howling wild, and strength was imparted to him as to a deliverer. He bore along the fair child in his arms, even as if she had been a lamb. The snow drift blew not—the wind fell dead—a sort of glimmer, like that of an upbreking and departing storm, gathered about him—his dogs barked and jumped, and burrowed joyfully in the snow—and the youth, strong in sudden hope, exclaimed, "With the blessing of God, who has not deserted us in our sore distress, will I carry thee, Hannah, in my arms, and lay thee down alive in the house of thy father."—At this moment there were no stars in Heaven, but she opened her dim blue eyes upon him on whose bosom she was unconsciously lying, and said, as in a dream, "Send the riband that ties up my hair, as a keepsake to William Grieve." "She thinks that she is on her death bed, and forgets not the son of her master. It is the voice of God that tells me she will not now die, and that, under His grace, I shall be her deliverer."

The short lived rage of the storm was soon over, and William could attend to the beloved being on his bosom. The warmth of his heart seemed to infuse life into her's; and as he gently placed her feet on the snow, till he muffled her up in his plaid, as well as in her own, she made an effort to stand, and with extreme perplexity and bewilderment, faintly inquired, where she was, and what fearful catastrophe had befallen them? She was, however, too weak to walk; and as her young master carried her along, she murmured, "O William! what if my father be in the moor?—For if you, who need care so little about me, have come hither, as I suppose to save my life, you may be sure that my father sat not within doors during the storm." As she spoke it was calm below, but the wind was still alive in the upper air, and cloud, rack, mist, and sleet, were all driving about in the sky. Out shone for a moment the pallid and ghostly moon, through a rent in the gloom, and by that uncertain light, came staggering forward the figure of a man.—"Father—Father," cried Hannah—and his gray hairs were already on her cheek. The barking of the dogs and the shouting of the young shepherd had struck his ear, as the sleep of death was stealing over him, and with the last effort of benumbed nature, he had roused himself from that fatal torpor and prest through the snow wreath that had separated him from his child. As yet they knew not of the danger each had endured—but each judged of the other's suffering from their own, and father and daughter regarded one another as creatures rescued, and hardly yet rescued from death.

But a few minutes ago, and the three human beings who loved each other so well, and now feared not to cross the Moor in safety, were, as they thought, on their death beds. Deliverance now shone upon them all like a gentle fire, dispelling that pleasant but deadly drowsiness; and the old man was soon able to assist William Grieve in leading Hannah along through the snow. Her colour and her warmth returned, and her lover—for so might he well now be called—felt her heart gently beating against her side. Filled as that heart was with gratitude to God, joy in her deliverance, love to her father, and purest affection for her master's son, never before had the innocent maiden known what was happiness—and never more was she to forget it. The night was now almost calm, and fast returning to its former beauty—when the party saw the first twinkle of the fire through the low window of the Cottage of the Moor. They soon were at the garden gate—and to relieve the heart of the wife and mother within, they talked loudly and cheerfully—naming each other familiarly, and laughing between, like persons who had known neither danger nor distress.

No voice answered from within—no footsteps came to the door which stood open, as when the father had left it in his fear, and now he thought with affright, that his wife, feeble as she was, had been unable to support the loneliness, and had followed him out into the night, never to be brought home alive.—As they bore Hannah into the house, his fear gave way to worse, for there upon the hard clay floor lay the mother upon her face, as if murdered by some savage blow. She was in the same deadly swoon into which she had fallen on her husband's departure, three hours before. The old man raised her up, and her pulse was still—so was her heart—her face pale and sunken—and her body cold as ice. "I have recovered a daughter," said the old man, "but I have lost a wife;" and he carried her, with a groan, to the bed on which he laid her lifeless body. The sight was too much for Hannah, worn out as she was, and who had hitherto been able to support herself in the delightful expectation of gladdening her mother's heart by her safe arrival. She, too, now swooned away, and, as she was placed on the bed beside her mother, it seemed, indeed that death, disappointed of his prey on the wild moor, had seized it in the cottage, and by the fire-side. The husband knelt down by the bed-side, and held his wife's icy hand in his, while William Grieve appalled, and awe-stricken, hung over his Hannah, and inwardly implored God that the night's wild adventure might not have so ghastly an end. But Hannah's young heart soon began once more to beat

—and soon as she came to her recollection, she rose up with a face whiter than ashes, and free from all smiles, as if none had ever played there, and joined her father and young master in their efforts to restore her mother to life.

It was the mercy of God that had struck her down to the earth, insensible to the shrieking winds, and the fears that would otherwise have killed her. Three hours of that wild storm had passed over her head, and she heard nothing more than if she had been asleep in a breathless night of the summer dew. Not even a dream had touched her brain, and when she opened her eyes which, as she thought had been but a moment shut, she had scarcely time to recal to her recollection the image of her husband rushing out into the storm, and of a daughter therein lost, till she beheld that very husband kneeling tenderly by her bed-side, and that very daughter smoothing the pillow on which her aching temples reclined. But she knew from the white steadfast countenances before her that there had been tribulation and deliverance, and she looked on the beloved beings ministering by her bed, as more fearfully dear to her from the unimagined danger from which she felt assured they had been rescued by the arm of the Almighty.

There is little need to speak of returning recollection, and returning strength. They had all now power to weep, and power to pray. The Bible had been lying in its place ready for worship—and the father read aloud that chapter in which is narrated our Saviour's act of miraculous power, by which he saved Peter from the sea. Soon as the solemn thoughts awakened by that act of mercy so similar to that which had rescued themselves from death had subsided, and they had all risen up from prayer, they gathered themselves in gratitude round the little table which had stood so many hours spread—and exhausted nature was strengthened and restored by a frugal and simple meal partaken of in silent thankfulness.

The whole story of the night was then calmly recited—and when the mother heard how the stripling had followed her sweet Hannah into the storm, and borne her in his arms through a hundred drifted heaps—and then looked upon her in her pride, so young, so innocent, and so beautiful, she knew, that were the child indeed to become an orphan, there was one, who, if there was either trust in nature, or truth in religion, would guard and cherish her all the days of her life.

It was not nine o'clock when the storm came down from Glen Scrae upon the Black-moss, and now in a pause of silence the clock struck twelve. Within these three hours William and Hannah had led a life of trouble and of joy, that had enlarged and kindled their hearts within them—and they felt that henceforth they were to live wholly for each other's sakes. His love was the proud and exulting love of a deliverer, who, under Providence, had saved from the frost and the snow the innocence and the beauty of which his young passionate heart had been so desperately enamoured—and he now thought of his own Hannah Lee ever more moving about in his father's house, not as a servant, but as a daughter—and when some few happy years had gone by, his own most beautiful and most loving wife. The innocent maiden still called him her young master—but was not ashamed of the holy affection which she now knew that she had long felt for the fearless youth on whose bosom she had thought herself dying in that cold and miserable moor. Her heart leapt within her when she heard her parents bless him by his name—and when he took her hand into his before them, and vowed before that Power who had that night saved them from the snow, that Hannah Lee should ere long be his wedded wife—she wept and sobbed as if her heart would break in a fit of strange and insupportable happiness.

The young shepherd rose to bid them farewell—"my father will think I am lost," said he, with a grave smile, "and my Hannah's mother knows what it is to fear for a child." So nothing was said to detain him, and the family went with him to the door. The skies smiled serenely as if a storm had never swept before the stars—the moon was sinking from her meridian, but in cloudless splendour—and the hollow of the hills was hushed as that of heaven. Danger there was none over the placid night-scene—the happy youth soon crost the Black-moss, now perfectly still—and, perhaps, just as he was passing, with a shudder of gratitude, the very spot where his sweet Hannah Lee had so nearly perished, she was lying down to sleep in her innocence, or dreaming of one now dearer to her than all on earth but her parents.

EREMUS.

MY NEIGHBOUR EPHRAIM.

I went this afternoon to pay a visit to my neighbour Ephraim; indeed I find his cheerful fire-side so much more pleasant than my own little solitary dwelling, that I am afraid I go there rather too often; however, as yet I have not remarked any coldness or distance in their reception of me. Ephraim had been a little indisposed, and I found him reclining on the sofa; his wife was preparing something comfortable for him by the fire, and his daughter, having arranged his pillow to his mind, sat with her work at his feet, while Ezekiel read to him—his other son was engaged in superintending the business of the farm; but when the hour of tea approached, he joined the circle in the parlour with a smiling countenance, cheeks glowing with health, and an appetite which appeared in no wise diminished by the exercise of the day. When I returned to my own lonely habitation, I could not avoid contrasting a little my situation with that of my old friend. Happy Ephraim! said I, thou hast an excellent wife and dutiful daughter, to smooth the pillow of thy aching head, to hover with feathery footsteps around thy peaceful couch, and watch over thy slumbers with the assiduity of anxious love—thou hast two manly intelligent sons, to attend to thy business, to protect thy interests, and support thy tottering steps; whose only strife is that of kindness, whose only rivalship, which shall be most attentive to thee, each of whom would gladly say with the poet,

Me, may the gentle office long engage
To rock the cradle of reposing age.—

And when at last, in a good old age, thou shalt be gathered to thy fathers, a train of mourning relatives shall deposit with decent care thy cherished remains in the narrow house appointed for all living; while I stand alone in the world, an insulated, insignificant being, for whom no one feels an interest, and whose pains and pleasures are of consequence to no one; whose approach is greeted with no smile, and whose departure excites no regret, and when the closing scene approaches, no kindred hand shall support my throbbing temples, or prepare the potion for my feverish lip, but

mercenary eyes alone mark with ill disguised impatience the uncertain flutter of the lingering pulse, mercenary attendants only receive, with frigid indifference, the last farewell of the departing spirit—

"By strangers' hands my dying eyes be clos'd,
"By strangers' hands my lifeless limbs compos'd."

Lost in a train of such like melancholy musings, and pondering on the past, the present and the future, I had suffered my fire to become nearly extinguished, and the feeble glimmer of my untrimmed taper faintly illumined my little study, when I was roused from my revery by the entrance of Ezekiel and his sister: The good girl said she had remarked that I was more silent than usual, and as the evening was fine, they had come over to see if I was unwell: this little act of kindness, though in itself no way remarkable, yet coming at such a moment, affected me not a little.— But I must shake off this gloom and depression of spirit, I am not now to learn that the world had much rather laugh *with* or *at* a man than mourn with him; I did not sit down to lament the desolation of my own situation, which cannot now be remedied; but to exhort the young to get married, to encourage them by the example of Ephraim, and to warn them of my own: "Do nothing in a hurry," is an excellent maxim in the main, but in some cases it is possible to use too much deliberation; in the important business of taking a wife, many a man has debated and deliberated, until the season for acting has passed away. An old fellow like myself has little to do in the world, but to talk for the benefit of his neighbours; and I would willingly devote my experience to the service of the rising generation. I should feel no objection to narrate the disastrous consequences of my own superabundant caution in the affair of matrimony, and to enumerate the many eligible matches which have slipped through my fingers; the opportunities to form advantageous connexions which have been unimproved in consequence of my hesitation and indecision, for I have now no plans to be defeated or prospects blasted by a knowledge of my failings, and no vanity to be mortified by the exposure of my disappointments, but I am apprehensive the detail might prove rather tedious and uninteresting; I may, however, mention a few circumstances attending my last attempts to obtain a help mate, if attempt it may be called. I had become acquainted in

the family of a respectable farmer, who had a daughter of a suitable age, and although I cannot say that

"Grace was in all her steps, heaven in her eye,
"In every motion, dignity and love,"

yet her correct and orderly deportment seemed to promise that she would make an excellent wife; I was therefore pretty frequent in my visits, and though on these occasions my discourse was principally, if not entirely, addressed to the parents, yet I kept a sharp eye upon the daughter, in order to endeavour to form a tolerable estimate of her disposition and character; and as I had in those days a handsome little estate at my own disposal, and was upon the whole considered rather a promising young man, my company seemed always very acceptable, to the father and mother at least.

In this manner, eight or ten months, perhaps, passed pleasantly away, and I was beginning to think that I might before long venture to address her with a little freedom and familiarity, preparatory to a serious negotiation when all my plans were defeated, and my visionary castle crumbled into dust, by the precipitation of others.

One evening I was sitting with them as usual, when after a little time the father and mother, on some occasion, absented themselves from the room, and left the daughter and myself together. As I had not the most distant suspicion that there was any design in their movements, and expected their return every moment, I took up the almanack, (being fond of reading) and had just got cleverly through it, when they returned. I thought I remarked something particularly scrutinizing in the looks of the mother, but I believe she soon discovered that I had done nothing but read the almanack. On my next visit, I felt no small trepidation, having a strong suspicion of what might occur; and, in fact, we were again soon left alone together—and now the consciousness of what was expected, kept me as silent as ignorance had done before. In my distress I looked about for the almanack, but they had taken it away. In vain I endeavoured to find something to say, my faculties seemed spell bound; and I sat, I know not how long, in a pitiable state of confusion and embarrassment, until my companion made some remark respecting the weather—this was a great relief. I immediately proceeded to treat of the weather in all its bearings, past, present and to come, and strove

to prolong the discussion until some one might come in, but in vain—the subject at length became exhausted, and silence again took place; which lasted so long, and became so glaringly ridiculous, that in utter despair, I was upon the point of having recourse to the weather again, when we were relieved by the entrance of company.

Determined never again to cut so silly a figure, I resolved to provide against my next visit a fund of agreeable conversation. I accordingly brushed up my acquaintance with the philosophy of Aristotle, and of the peripatetics generally; collected some anecdotes of the wise men of Greece, and, not to lack matters of more recent date, stored my memory with a few amusing particulars respecting Mary Queen of Scots, and of the court of Elizabeth: thus prepared, I ventured once more to make my appearance, but I had no opportunity to say a word about Aristotle or the Queen of Scots; it was rather late when I entered the room, and I found my intended in earnest conversation with a young man who had drawn his chair very near to her: their discourse seemed to be of an interesting nature, but they spoke in so low a tone, that I was unable to profit by their remarks; I observed at last, that they frequently smiled when looking towards me, and as I love a cheerful countenance, and smiling is certainly contagious, I smiled a good deal too. This seemed wonderfully to promote their risibility, and my laughter increasing in the same proportion, we had a deal of merriment, although little or nothing was said: how long this might have continued I know not, had not my intended father-in-law called me aside, and hinted that as the night was dark, and there was some appearance of rain, I had perhaps better return. I thanked him for his truly paternal care, and accordingly took my departure in high good humour, and the next week was informed that the young people were married.

[*Rural Visiter.*

VIEW OF A GOSSIP.

Mr. Editor—I send you the following account of a short inspection of a fellow creature, which, if it will convey any information to your readers, you are at liberty to publish.

I had the curiosity, as a neighbouring gossip was one day at my house, and while she happened to be napping on the sofa, to try if I could obtain a view of her ear, the structure of which I had often speculated about. With the aid of a good eye-glass I succeeded. It was a curious piece of mechanism: the outside folding was of the usual size, but by long habit of eager listening, had acquired a kind of gaping shape, which seemed to bid an indiscriminate welcome to every sound. Next to this was a kind of whispering gallery, so extremely susceptible of noise, that one of my most careful breathings was immediately reverberated to the tympanum, and though not loud enough to awake the sleeper, it was evident that it made her dream of scandal. I was curious to know the construction of the tympanum, which vibrated so easily to an empty breath. I could see it plainly: for the long habit of extreme anxiety to receive every breath that stirred, had pushed it forward to the very orifice of the ear, where it seemed waiting in an agony of impatience for something new, and complaining grievously that all the tit-bits of intelligence had to pass such a distance before they could come to its hearing. Its make was curious: it was an extremely fine sieve of the most elastic materials. It was evidently constructed so that every thing of the least weight should rebound as soon as it touched, and only those articles which were as light as air, should enter. It was too fine to admit any thing larger than a bit of scandal. The whole external ear was designed to catch and communicate every thing audible, for it is a maxim with gossips as well as others, that nothing is too poor for a nice hand to sift some good from it. This sifting was the office of the tympanum: it was in a perpetual quiver. Its nice threads were constantly shaking to pieces what was too light to rebound, and too large to enter whole, and then dropping them to the receiver below. There were a great many curious articles sticking in this sieve, which had got half through but could go no further. There were little pieces of serious and affecting family secrets, tales of distress, cullings from

the little failings of good men, morsels of sermons, drippings of church business, ends, middles and halves of people's sayings, anecdotes of funerals, with numerous suspicions, half heard hear-says, and suspicions of near-says.—All these had evidently been operated upon. A kind of scandalic acid had been at work on those parts which had got within. They were partly decomposed; their seriousness had assumed a ridiculous aspect, their solidity had become gaseous: what had been affecting, seemed now to be laughable; what had appeared commendable, seemed now composed of so many disgusting materials as to be odious. Here I learned the reason which I never knew before, why it is that gossips hear so many lectures upon the degrading, injurious, disreputable character of their pursuits, and so entirely without effect. These lectures never passed their ears: their matter was too heavy not to rebound, or the truths too great to gain admission; or, if any detached parts chanced to enter, they were so broken by sifting, and so decomposed and changed by the very pungent acid within, as to retain nothing of their original seriousness, and become fit companions for the nice selections which passed before them.

The sight of the ear excited my curiosity to look for the mind within: whatever may be the difficulty of determining the seat of the mind in persons of ordinary construction, the matter is clear, that the gossip's mind must be as near to the vehicle of sound as possible. This is confirmed by observation; I found it just inside the ear, where there was but a small space, to be sure, for its operation; but its dimensions were of no enlarged description. It had evidently been made for some useful employment: its nerves were strong, its perceptions quick, its action skilful; but it was miserably contracted. Part of it, and plainly the best part, was ruined by long inaction. It contained a few good ideas, grown rusty by long disuse, which evidently might have been made to appear respectable, could they have been delivered from the stuff which covered them. This mind was a complete factory in miniature: there was its picking machine, its spinning machine, a contrivance for weaving, for shearing, for trimming; with dyes of every variety of shade. All these machines were so artfully combined, that you could see nothing of the raw material after it once went to be picked, till it came out in an article nicely dressed and dyed for distribution. There was a very smooth communication from the mind to the mouth, through which the different articles, as soon as finished, were conveyed, to be *kindly*, and *complacently*, and *charitably*, rattled out to every one that

should come. A peculiar excellence of this mechanism was, that there seemed to be no refuse: every thing was worked, every thing was turned to some account: the motion here was perpetual, and acid was strangely used instead of oil to facilitate it. All around, there were receipts for the best method of making a good story out of almost nothing; of how to extract something laughable from the most serious subjects; of how to dye white into a good black, and how to find materials for manufacturing, where no one would think of looking. The day-book was a curiosity. "April 1st, eked out of John B's apprentice two skeins of scandal about his master." "Monday, learned a good deal from Mrs. C's cook about her mistress' private ways, and gained a variety of nice bits, which, with a chain of good *home-spun*, will make up a very good article." "Tuesday, heard a whisper about something disagreeable in Mr. D's family, can't rest till I know more about it; must send Sally into their kitchen to see what can be picked up there. A few family quarrels would help finely just now." "Wednesday, caught the thread to the tale which I have been trying these three months to unwind, it seems to lead to some noble pickings." "To-day, must go out and hear the news, and ask about this marriage rumour."

As I was finishing my observations, a little neglected thing struck my attention, which seemed to demand some notice. It was the gossip's conscience. It was a little, contracted, fantastical body, that seemed extremely averse to noise and all kinds of disturbance; to avoid which, it had squeezed itself into a narrow corner, where, with the help of several ingenious contrivances, it kept clear of all interference: it seemed to sleep almost all the time, in which it was assisted by the influence of numerous little nostrums which were kept for the purpose. When conscience did sometimes awake, (as I conjectured) it was mostly on a Sunday, while there was but little doing, *comparatively*, when it looked about a little, made some bustle, and went to sleep again perfectly satisfied. From appearance, I judged that conscience and the gossip had very little intercourse.

One thing was remarkable about this mental factory, which was, the flourishing state of its business; a circumstance which seems the more strange, because such establishments are so very numerous, and their productions so eminently worthless.

Now, sir, having finished my description, permit me to ask, if there is any thing to excuse the employment of gossips, great or small? Can the want of

other occupation, or the amusement which this affords them, make amends for such degradation of themselves; such abominable trifling with their neighbour's character; such vexatious meddling with other's business; such remorseless transformation of good into evil; of secrets into public news; of the serious into the ridiculous; of peace into disputing; as they are constantly guilty of? Ought not such persons to be universally shunned as public evils, and if a public law will do no good, should it not be the secret resolution of every gossip-hater, to avoid as a pestilence, the scandalous atmosphere of a scandalous tale-bearer?

There is a celebrated description of law which affords a good outline for the description of what of all things is most lawless. Of scandal, there can be no less acknowledged, than that her seat is in the temple of fame; her voice the confusion of the world; all things in earth and hell own her influence; the very least as feeling her hate, and the greatest as not exempt from her power; both men and women, and creatures of what condition soever, though each in different sort and manner, yet all with uniform consent, detesting her as the pest of their peace and joy.

ANTI-TATTLER.

To prevent Skippers in Bacon.—Take of red pepper finely powdered one spoonful for every joint of meat, and rub it on the meat with the salt, when it is first cut up. It has been often tried and was never known to fail in producing the above effect.

A DISCOURSE, READ BEFORE THE
Essex Agricultural Society,
In Massachusetts, February 21, 1820,
*Suggesting some Improvements in the Agriculture
of the County.*

BY TIMOTHY PICKERING,
President of the Society.

At a Meeting of the Essex Agricultural Society, at Topsfield, February 21,
1820,

Voted, That the thanks of the Society be presented to the Hon. TIMOTHY PICKERING for his interesting Address, and that he be requested to furnish a copy thereof for publication.—Attest,

FREDERICK HOWES, *Secretary*.

DISCOURSE.

Gentlemen—The secretary has put into my hands a vote of the Society, requesting me to "make to it such communications as may in my opinion most conduce to the interest of Agriculture."

This was an unlooked for request. I have myself much to learn from observing farmers, of longer experience, and whose attentions have been exclusively devoted to husbandry. Mine, since I became a farmer, have been diverted by other pursuits; so that at intervals only my thoughts have been turned to this subject.

No one doubts the importance of our profession; and the actual formation of our society is a declaration that improvements in it are necessary.—But the field of agriculture is of boundless extent; and though traversed for some thousands of years by the greater portion of the human race, yet by no one, nor by all combined, has a complete survey been accomplished. Every year, and every day, presents something new: and even of old things, the practices of ages, there still exist diversities of opinions. For instance, which is preferable, deep or shallow ploughing?—Should manures be spread on the surface, or be buried by the plough? If the latter, at what depth, to produce the greatest effect, with the most lasting fertility?—Should manure be applied in its rough, coarse, and unfermented state, or, by keeping and repeated turnings, be more or less rotted?—These are points which appear to me deeply to affect the interests of agriculture. On these therefore I will give you my opinion, enlightened by the observations of intelligent husbandmen. I will then advert to a few other topics which demand your attention; *dwelling* on one of them—Root crops for the Food of Live Stock—as lying at the foundation of an improved agriculture.

I. *On Deep Ploughing and Manuring.*

For myself, I entertain no doubt of the utility of deep ploughing; not at once, in our lands in general, but by an increase of two or three inches at every annual ploughing, until the earth be stirred and pulverized to the depth of ten or twelve inches. Indian corn, planted in such a mass of loosened earth, would not, I am persuaded, ever suffer by ordinary

droughts.—Like a sponge, it would absorb a vast quantity of rain water, and become a reservoir to supply the wants of that and all other plants.—Nothing is more common in a dry summer, than the rolling of the leaves of corn; and that circumstance is often mentioned as an evidence of the severity of the drought. This rolling of the leaves of Indian corn, is the consequence, in part, of scant manuring, but still more of shallow ploughing. Few, perhaps, are aware of the depth to which the roots of plants will penetrate in a deeply loosened earth. A gentleman,^[11] much inclined to agricultural inquiries and observations, informed me, near fifty years ago, that seeing some men digging a well, in a hollow place, planted with Indian corn, then at its full growth, he stopped to examine how far its roots had descended; and he traced them to the depth of nine feet. The soil was an accumulation of rich earth which had run or been thrown into the hollow.

The seed of the common turnip, sown in warm weather, and on a soil sufficiently moist, I have known to vegetate in about eight-and-forty hours; and in only four or five days afterwards, I found the plants had sent down roots to the depth of four or five inches.

I have often noticed forest trees blown down by violent winds, whose roots, of the same species, were very differently formed. Such as had grown in grounds having a hard, impenetrable pan of clayey gravel, at the depth of twelve or eighteen inches from the surface, exhibiting a flat mass of roots; while others, torn up from a deep loam, or loamy gravel, showed downward roots of several feet in length.

About five months ago, I received from England a pamphlet written by one of the most distinguished agricultural writers in that country—Arthur Young. It was a lecture read, a few years before, to the British Board of Agriculture, of which Mr. Young was the Secretary. Its title is, "On the Husbandry of three celebrated British Farmers, Messrs. Bakewell, Arbuthnot and Duckett,"—all eminent for genius, enterprise, application, and long experience. It was to do honour to their memories, "and to bring to recollection the means by which those celebrated practitioners, in the first and most important of all arts, carried their agriculture to a perfection unknown before," that the lecture was written and published. And this, Mr. Young observes, would be more peculiarly useful, because those men, "confining themselves to practice alone, had left no register of their own meritorious deeds." I will present to you the substance of the information

contained in this pamphlet, as in itself very important, and because the practice of Arbuthnot and Duckett has a direct bearing on the points I am now considering—DEEP PLOUGHING and MANURING.

"Mr. Duckett had sand, and sandy soils alone, to deal with; but Arbuthnot's land classed among those harsh, wet, tenacious loams, which are usually called clay, and ought to be esteemed such, relative to every circumstance that attaches to difficulty and management." Passing by what Mr. Young says of Arbuthnot's draining operations, I content myself with mentioning the principal of that improvement: "Lay your land dry, whatever may be the method pursued, before you attempt any thing else."

"In respect to tillage, Mr. Arbuthnot carried it to great perfection: He invented a swing plough for a pair of horses and the general depth of six inches, and a much larger one with wheels, for gaining the depth of 12, and even of eighteen, for some peculiar crops, especially madder. Upon the advantages of deep ploughing he never had the least hesitation; but always declared that in all he had read or heard, he never met with one argument against the practice that had with him the smallest weight."—"In the essential operation of ploughing, he considered one earth [that is, one ploughing] well timed, and of a right depth, as being much more efficacious than that repetition of tillage so common in every district."^[12]

A judicious rotation, or round of crops, has long been considered, in England, essential to good husbandry: and so it is by skilful farmers in our own country; particularly in the middle States, where clover, so highly important in the rotation, has, for more than thirty years, been rendered wonderfully productive, by the application of plaster of Paris. The most usual course in England has been (excepting on stiff clayey soils) first year turnips, manured and kept clean by hoeing; the second year barley, with clover seed; the third year the clover mown for hay; and its second crop, at wheat seed time, ploughed in, and, where necessary to fill the seams, the ground harrowed, the wheat sown, and then harrowed in. This is called "wheat upon a clover lay."—But by the long and frequent repetition of clover, (that is, once in *four* years) in their rotations, lands in England became (as they express it) "sick of clover:" and I have been informed that some lands in our middle States, long subjected to the like application of clover, exhibit like symptoms of disease or failure. But Mr. Arbuthnot introduced clover once in *three* years, without suffering by such more

frequent repetition. "He attributed the failure of this plant to shallow and ill-executed ploughing; the result (says Mr. Young) justified his opinion."

Mr. Young mentions a lecture he had read to the Board of Agriculture, "on the means by which a farm can be made, by a right proportion of all the products, to support itself, without foreign assistance, in a state of high fertility, a question depending on the quantity or weight of dung resulting from the consumption in litter of a given weight of straw." This lecture I have not seen. But he considers the question as successfully decided, in Mr. Arbuthnot's practice, in the following manner; 134 sheep and 30 lambs were turnip fed, in a pen on a headland, well littered with straw: in six weeks they required nearly six tons of straw [to give them clean and comfortable beds:] and in that time made 40 tons of dung, equal to that brought from London [stable dung it is to be presumed.] So every ton of litter produced near seven tons of dung.—But this weight must have been obtained chiefly by the earth of the headland absorbing the urine, of which, when fed on turnips, sheep make great quantities, and being finally mixed with their dung and litter. This recital reminds me of the recommendation, in my address to this Society, in May, 1818, to carry earth into the barn yard, once in every two weeks, from spring to autumn; adding to every layer of earth a coat of litter. I should then have advised a plentiful spreading of litter, had I not known that our courses of husbandry in Essex yielded very little straw.

In the same communication to the Society, I presented my ideas on the proper application of manure; to wit, *always to bury it up quickly, when carried to the field*, to prevent great loss by its exposure to the sun and air; remarking, that the essence of manure was lost, not by *sinking* into the earth below the roots of cultivated plants, but by *rising* into the atmosphere, and so fleeing away. Here, also, I have the satisfaction of seeing the theory I had formed nineteen years ago (in the manner suggested in that communication) supported by the opinions and practices of such eminent agriculturists as Messrs. Arbuthnot and Duckett. After noticing Arbuthnot's cultivation of madder, an article requiring a rich soil and extremely deep tillage, Mr. Young says, "there was one circumstance in his management, which, being applicable to more important articles, merits a more durable attention; this is, the depth to which he ploughed in the dung: his tillage went to that of eighteen inches; and he conceived there was no danger of losing, by this

circumstance, either *vegetable* or *animal* manures, as *their* tendency, contrary to all *fossil* ones, was not to *sink*, but to *rise* in the atmosphere." Fossil manures are lime, marl, plaster of Paris, and other substances dug out of the earth, which increase the productive powers of soils.

Mr. Duckett's manner of applying dung, although his was a sand farm, was similar to Mr. Arbuthnot's.—"Immediately connected with the depth of tillage, is that to which dung may be safely deposited. He [Mr. Duckett] had not the least apprehension of losing it by deep ploughing; but freely turned it down to two or three times the depth common among his neighbours." Yet Mr. Young says, that farmers (and good farmers too) persist in a contrary practice. But he adds, "Enlightened individuals, thinly scattered, know better: having convinced themselves that Mr. Duckett's practice is not only safe but beneficial;" and then names one who "ploughs in his dung as deeply as his ploughs can go, turning it in nine inches, and would bury it twelve, did he stir to such a depth."

Confirmatory of the correctness of the practice of these two celebrated English farmers, is the fact stated by Mr. John Sinclair, President of the British Board of Agriculture, in his account of the Improved Scottish Husbandry. He mentions one farmer who ridged his carrot ground, and buried the manure sixteen or seventeen inches deep, the ridges thirty inches wide. This farmer preferred, as a manure, a well prepared compost of peat-moss^[13] and dung, ten tons, or double cart-loads, per English acre. "The dung (or compost) being at the bottom, makes the tap root of the carrot push immediately down, and swell to an enormous size; the roots being often sixteen inches in girth, and 18 or twenty inches in length."

To return to Mr. Duckett. His deep ploughing (says Mr. Young) was not practised above once in two or three years, and the successive tillage shallow. "By such deep ploughing, seldom given, Mr. Duckett conceived that a due degree of moisture was preserved in his *light* land, by means of which his crops were flourishing in seasons of drought which destroyed those of his neighbours: and no one could more severely condemn the ideas which governed the Norfolk farmers, in leaving what they called their *pan* unbroken at the depth only of 4 or 5 inches.—The operation of ploughing he thought could scarcely be given too seldom, provided when given it was done effectively: and he always carried this paucity of tillage as far as circumstances would permit: thus I have known him put in seven crops with

only four ploughings." In another part of his lecture, Mr. Young says, "If I were to name the circumstance which more than any other governed his (Mr. Duckett's) practice, I should say that the whole was founded in trench ploughing; and that the principle which governed this practice (a principle thoroughly impressed upon his mind, as well as on the minds of those who draw intelligent conclusions) was that of giving as little tillage as possible to sandy soils."

"The next circumstance which I shall advert to (says Mr. Young) in the husbandry of Mr. Duckett, is the use of *long, fresh dung*, instead of that which in common management is turned and mixed till it becomes *rotten*: and in justice to his memory, I shall read the short recital of his practice, as I printed it three-and-twenty years ago. "Dependent on the Trench-Plough, ^[14] is Mr. Duckett's system of dunging. He conceives, and I apprehend very justly, that the more dunghills are stirred and turned over, and rotted, the more of their virtue is lost. It is not a question of straw merely wetted; but good *long* dung he esteems more than that quantity of *short* dung, which time will convert the former to. *Two* loads of *long* may become *one* of *short*; but the two are much more valuable than the one. Without the Trenching-plough, however, his opinion would be different. If long dung is ploughed in, in the common manner, with lumps and bundles sticking out at many places along every furrow, which lets the sun and air into the rest that seems covered, he thinks, so used, it is mostly lost, or given to the winds: in such a case, short rotted manure will be better covered, and should be preferred. But with his plough nothing of this happens; and it enables him to use his dung in such a state as gives him a large quantity instead of a small one. The good sense of these observations must be obvious at the first blush." Mr. Young adds—"The use of FRESH instead of ROTTEN dung, is, in my opinion, one of the greatest agricultural discoveries that has been made in the present age." He then states a striking experiment made by himself—67 small cart loads of fresh yard dung produced two successive crops of potatoes, yielding together 742 bushels; at the same time, the same quantity of yard dung, after 6 months rotting, yielded 708 bushels, leaving [to the fresh long dung] a superiority of 34 bushels. But had the fresh dung been kept as long as the other, it would have required at least twice, perhaps thrice as much, to have produced the quantity used." [That is, twice or three times 67 loads of fresh long dung, if kept and often turned and mixed to produce fermentation and rotting, would have shrunk, or been reduced, to

67 loads of short rotten dung.] "If the crops therefore had been only equal, still the advantage [of the fresh dung] would have been most decisive."

"I shall not quit (says Mr. Young) the husbandry of two men who carried tillage, on soils so extremely different, to its utmost perfection, without remarking the circumstances in which they agreed. Both were equal friends to deep ploughing; both rejected the common repetition of tillage, and reduced the number of their operations to a degree that merits attention; both rejected fallows; and both ploughed deeply for depositing manure, without any apprehension of losing it. These are very important points in Practical Agriculture."

To this account of the successful practices of these two celebrated English farmers, it may be useful to subjoin a few observations. I have thought it proper so far to present them in detail, in order to develop principles; not expecting a *precise* adoption of their practices; which indeed, without their or similar superior ploughs and other implements, would be impracticable: but with such instruments as we possess, or may easily obtain, we can materially increase the depth of our ploughing, and I hope contrive effectually to cover our manure. *This should be wholly applied to Tillage Crops*; for which the manuring should be so ample as to ensure a succession of good crops through the whole rotation, without the aid of any additional manure, especially for wheat, rye, barley or oats: for besides increasing the seeds of weeds (with which all our lands are too much infested) such additional manuring, *immediately* applied to the small grain crops, renders them more liable to injury from mildews. Of this I am fully satisfied, as well from numerous statements of facts which I have seen in books of husbandry, as from the circumstances under which remarkable mildews have otherwise been noticed. One of our countrymen, who wrote a short essay on the subject prior to the American Revolution, has given the only solution of the causes of mildews that has ever appeared satisfactory to me: perhaps at some future time I may find leisure to show the correspondence of facts with his principles.^[15]

(To be continued.)

To Farmers.—In the winter of 1818-19, a gentleman in this city made the following experiment. He placed a turkey in an enclosure about four or five feet long, two feet wide, and three or four feet high. He excluded as much light as he could without preventing a circulation of air, and fed the turkey with soft brick broken into pieces, with charcoal also broken, and with six grains of corn per day. Fresh water was daily supplied. The box or coop in which the turkey was placed, he always locked up with his own hands, and is perfectly confident that nobody interfered with the experiment.

At the end of one month he invited a number of his neighbours among others two physicians. The turkey, now very large and heavy, was killed and opened by the physicians, and was found to be filled up full with fat. The gizzard and entrails were dissected, and nothing was found but a residuum of charcoal and brick. To conclude the examination satisfactorily, the turkey was eaten, and found to be very good.

Last winter he again repeated the experiment with the same success.

The circumstance by which he was induced to make the experiment is a very curious one. One of his neighbours informed him, that being driven from the city by the fever of 1793, his family recollected that some fowls that had lived in a kind of loft over his workshop, had been forgotten in the hurry of their removal, and would certainly be starved. They were absent six or eight weeks, and on the retiring of the pestilence returned. To their great astonishment, the fowls were not only alive, but very fat, although there was *nothing but charcoal and shavings* that they could have eaten, and some water that had been left in the trough of a grindstone had supplied them with drink.

[*Nat. Recorder.*

Introduction of Glass Making in France.

(From Parke's Chymical Essays.)

The government of France, in the early part of the fourteenth century, took great pains to improve the manufacture of glass, and ordained that none but gentlemen, or the sons of the nobility, should be allowed to exercise the trade, or even to work as artificers in the manufactories of this most highly esteemed commodity.—In consequence of this injunction, a company of persons, all born gentlemen, was incorporated, and obtained many important privileges and immunities from the state; particularly that of being allowed to work at this curious art without derogating from their nobility. It is indeed asserted by the writer,^[16] who is the best authority we have on this subject, that there never was an instance of any one being attainted, to whom these privileges had been granted; for they conducted themselves so irreproachably, that these orders were invariably transmitted inviolate to their posterity. In the year 1453, Anthony de Brossord, Lord of St. Martin, and prince of the blood royal, finding the business of glass making to be so considerable; and knowing that it did not derogate from nobility, obtained a grant from the Prince to establish a glass house in his own county, with prohibition of any other; and in consequence of this, the elder sons of that family continued uninterruptedly to exercise the art till the latter end of the sixteenth century, when the proprietor was killed while commanding a troop at the siege of Chartres.—On the death of this individual the younger sons of the same family undertook to carry on the art, and continued in it for more than a century. Whether the trade continues still in the same line, I have not been able to ascertain.

An ancient family of the name of *Vaillant*, also obtained the grant of a glass house, as a recompense for their valour and public services, together with a poignard d'or, on azure, for their arms.—Mr. Blancourt, who long resided in France, likewise notices, that at the time he wrote, they had many other great families among them, who were descended from gentleman glass-makers that had declined following the art; and that some of these had

been honoured with purple, and with the highest dignities and offices in the state.

FOR THE RURAL MAGAZINE.

Anecdote of Anthony Benezet,

Not inserted in Vaux's interesting Memoir of that Philanthropist.

Soon after the arrival of the Chevalier Luzerne, minister from the court of France, Benezet called on him with a French copy of Barclay's Apology, with a view of informing him of the principles of the Society of Friends. The minister being a Knight of Malta, and of course at enmity with the Turks, appeared much surprised that any professed Christians should object to the destruction of the Ottoman Empire—which increased on being informed that his profession would not permit taking the life of any man: on which the minister observed that it was *very good*, but too *straight* for him to object to killing a Musselman.

The interview prepared the way for frequent visits to the ambassador, who always received him with pleasure, the latter often observing that he had but a small body; but added, extending his arms nearly at full length, as if to embrace a large object—"*Oh what a capacious soul he possesses!*"—evincing by his whole conduct, that he valued him as an extraordinary man, possessing true Christian principles.

G.

MISCELLANY.

***The English, a Foreign Tongue.*—We extract the following from a French paper:**

"An etymologist has lately published the following analysis of the English language. Its vocabulary, he says, is composed of 6621 words of Latin origin, 4361 of French, 2068 of Saxon, 1288 of Greek, 660 of Dutch, 229 of Italian, 117 of German, 11 of Gaelic, 83 of Spanish, 81 of Danish, 18 of Arabic, besides many others of ancient Teutonic, Hebrew, Swedish, Portuguese, Flemish, Russian, Egyptian, Persian, Cimbrian, and Chinese!! The same etymologist pretends, that in Shakspeare, Pope, Swift, and Milton, *there are not many more than a hundred words purely English!*"

So that it would appear, that when we meet an acquaintance in the street, and accost him with "How do you do this morning?" and he replies "pretty well, I thank you," we are probably speaking half a dozen languages at once. What "learned Thebans" we must be! In this way a man who has a tolerable understanding of *Dilworth's spelling book* must be no inconsiderable linguist while one, who can read *Johnson at sight*, must be a perfect *Polyglot*. The poor Burgeois gentilhomme was quite amazed to find, that he had been speaking *prose* all his life without knowing it; and we are no less astonished on discovering that we had been talking *Russian, Egyptian, Persian, Cimbrian, Teutonic, and Chinese*, for years, without having ever dreamed of it. There have been great controversies among the learned as to what was that formidable dialect, which arose at Babel, out of the confusion of tongues, but after this discovery we can have no doubt that it is that very English which we all speak, and instead of High Dutch, which some have supposed was the language used by Adam in Paradise, we do now verily believe, that it was that *pure English*, of which so few traces have been left!—*Lou. Adv.*

"*A Stitch in time, saves nine.*"—A celebrated French writer on political economy, M Say, has this story:—"Being in the country, I had an example of one of those small losses which a family is exposed to through negligence. For the want of a latchet of small value, the wicket of a barn yard looking to the fields, was often left open; every one who went through, drew the door to; but having no means to fasten it, it remained flapping: the poultry

escaped, and were lost. One day a fine pig got out and ran into the woods. Immediately all the world was after it; the gardener, the cook, the dairy-maid, all run to recover the swine. The gardener got sight of him first,—and jumping over the ditch to stop him, he sprained his ankle, and was confined a fortnight to the house.—The cook, on her return, found all the linen she had left to dry by the fire, burned; and the dairy-maid having run off before she had tied up the cows, one of them broke the leg of a colt in the stable. The gardener's lost time was worth twenty crowns, valuing the pain at nothing; the linen burned, and the colt spoiled, were worth as much more. Here a loss of forty crowns and much pain, trouble, vexation, and inconvenience, for want of a latch, which would not cost three pence, and all through careless neglect.'

Rats.—The following curious mode of catching rats is extracted from the works of Muller, an apothecary of Weringerode, in Germany:—

Procure a large cask, and place it in the vicinity of places infested with rats. During the first week, this vessel is employed only to allure the rats to visit the solid top of the cask, by means of boards or planks arranged in a sloping direction to the floor, which are to be strewed with oatmeal daily, or any other food grateful to the palate.—Being thus lulled into security, and accustomed to find a regular supply for their meals, a skin of parchment is substituted for the wooden top of the cask, and the former is cut for several inches in the centre in transverse directions, so as to yield on the slightest pressure. At the same time, a few gallons of water, to the depth of six or seven inches, are poured into the empty cask, in the middle of which a brick or stone is placed, so as to project one or two inches above the fluid, and afford to one rat a place of refuge. These measures being taken, the top of the cask should be furnished with the proper baits, in order to induce the marauders to repeat their visits. No sooner does one plunge through the section of the parchment into the vessel, than it retreats to the brick or stone, and commences its lamentations for relief. Others follow, and share the same fate. A dreadful conflict then commences to obtain possession of the dry assylum. Battles follow in rapid succession, attended with such loud and noisy shrieks, that all the rats in the neighbourhood hasten to the fatal spot, where they experience similar disasters. Thus, hundreds may be caught by stratagem, which might be greatly facilitated by exposing a living rat taken in a trap, or purchased from a professional rat catcher.

Important to Millers.—A very valuable machine has lately been invented (and is now in operation) by Richard French, of Morrisville, (Pa.) for cleansing all kinds of grain and grass seeds. This machine cleanses the grain completely of the white cap, and at the same time rubs off the dust that always adheres to grain, and is the cause of specks in flour. The grain at the same time receives a fine polish. It operates equally on rye and buckwheat, cleansing it from all the dust and fuz which darken and cause grit in the flour, and are so liable to fret the bolting cloth. It is the opinion of a number of millers, that one barrel of flour may be obtained more from every hundred bushels of wheat, cleaned by this machine, than from the same quantity in the usual way. This machine will remain at Morrisville, for public inspection, a few weeks, after which it will be removed to Brandywine. Millers and others, who wish to make more and better flour, as there is no loss of grain, are invited to call and see the machine in operation. I believe they will not regret the time and trouble of so doing.

[*Trenton paper.*]

Rhode Island.—The thrifty little state of Rhode Island is, at this time, the most prosperous of any in the Union, notwithstanding the multitude of small banking institutions that abound there, from the force of domestic industry applied to manufactures; which, in despite of every obstacle, is in a condition that must be considered a happy one, compared with that of most other places; a most rigid economy, in some measure, supplying the want of public protection, except in the people themselves—who chiefly consume the products of their respective neighbourhoods.—The balance of trade is generally in favour of the state, and the want of specie is not felt by those who have a right to demand it, for the banks are in a very *comfortable* state.

Maine.—The first organization of the government of the state of Maine, took place on Wednesday, May 31st, at Portland. John Chandler was unanimously chosen Speaker of the Senate, and Benjamin Ames chosen Speaker of the House of Representatives, also unanimously. General William King elected governor, by 20,000 votes out of something over that number, was qualified next day.

Oil Stones.—Within a few weeks past a body of very superior oil-stones has been discovered in the neighbourhood of Easton, (Penn.) We believe they were first discovered on the farm of George Ibrie, Esq. on the river

Lehigh, in Williams township. The bed, however, is not confined to that farm alone, but extends to a considerable distance on each side of the Lehigh. They have heretofore been found lying on the surface of the earth, and it is matter of astonishment they were not sooner discovered.—The carpenters of that place have almost entirely substituted them for the Turkey stones heretofore used for setting tools; believing them to be superior to those of Turkey, and infinitely better than those gotten near Oley, which they somewhat resemble in colour.—The discovery is certainly valuable to the mechanics, as the Turkey stones have sold there for 75 or 100 cents, and the Oley stones at 25 cents per lb.

Longevity of the Land Tortoise.—An article dated Sunbury, (Penn.) June 15, says, that as a person was lately rolling logs near Shamokin creek, he discovered a land tortoise, and through curiosity picked it up, when the following engraving was observed upon the under shell:

"Thomas Musgrave, 1712;"

and immediately below, in large capitals,

"ROBERT HUNTER, 1790"—

the former having been engraved 108, and the latter 30 years ago. He engraved his own name below, and set the Recorder of ages at liberty.

Great crop.—A. Mr. Blakeman, at Silver creek, Indiana, has published that he raised last year 1350 bushels of shelled corn on *ten* acres of newly turned prairie land.

Western navigation.—The Louisiana Advertiser, of the 6th of May, gives the names and destination of *twenty-three* steam-boats then lying in the port of New Orleans!

Pennsylvania Hospital.—The expenditures of the Pennsylvania Hospital the last year were about 55,000 dollars. The number of patients during the year ending April 22, 1820, was 945. The deaths were 52. Persons relieved 81. Cured 428. Remain 209. Eloped 23.

Salt water.—The whole western country seems to be *under*-flowed with salt water. Some late borings through the rock, at depths of from 259 to 317 feet, at Cannonsburg, Pa. have been completely successful.

Fire at Savannah.—An advertisement from the London Phenix Fire Office states, that the loss occurring in Savannah, by the great fire of January last, swept away every thing that had been received for premiums during twelve years, and as much more.

British Revenue, &c.—The *ad valorem* duty, on British manufactured goods, exported from Liverpool, amounted, for the first quarter of the year 1819, to upwards of 13,000*l.* For the first quarter of 1820, the amount was only 5,700*l.* A great falling off, indeed.

Average price of grain in England and Wales, from the returns up to the 15th April—Wheat, 69s. 2d.; Rye, 41s. 9d.; Barley, 36s. 5d.—[Grain at these prices would afford a fine market for the surplus product of the United States; but England will not receive our bread stuffs; she prefers to keep up, and to a most unreasonable extent, the market for her own agriculturists.]

Origin of Almanacks.—The ancient Saxons used to engrave upon certain square sticks, about a foot in length, the courses of the *moons* for the whole year, whereby they could always certainly tell when the new moons, full moons, and changes, should happen; and such carved sticks they called *Al-mon-aght* (*all-mon-heed*, *i. e.*) the regard or observation of all the moons. There is in St. John's College, Cambridge, a Saxon almanack exactly answering to the above description.

Iron Boat.—A London paper of May 4, says, that a passage boat, of malleable iron, now plies on the Forth and Clyde Canal, in Scotland. It is called the Vulcan, and succeeds to admiration. The length is 63 feet; beam, 13 feet; depth, 5 feet; draught of water, when launched, 22 inches abaft, and 19 inches forward—when fitted with cabins, &c. 37 and 25 inches—when laden with two hundred passengers and their baggage, under 48 inches, on an even keel. The weight of iron employed was 12 tons 11 $\frac{3}{4}$ cwt. which is less than a wooden vessel of the same dimensions. The iron is of the kind called scrap.

A Great Eclipse of the Sun will take place on the 7th of September next. This eclipse will be visible over an extent of more than four millions square leagues, a surface nearly equal to a sixth part of the earth, and resembling a kind of oval of about 7500 leagues in circumference, comprising all Europe, the western part of Asia, all Africa, as far as to Monopotapa, and a part of North America. The eclipse will last three hours.

Lord Thurlow.—This eminent lawyer's superiority of ability was very early manifested both at school and at college. They extorted submission from his equals, and impressed his seniors with respect. The following anecdote is told of him.—Having been absent from chapel, or committed some other offence which came under the cognizance of the dean of the college, who, though a man of wit, was not remarkable for his learning. The dean set Thurlow, as a task, a paper in the Spectator to translate into Greek. This he performed extremely well, and in very little time; but instead of carrying it up to the dean, as he ought to have done, he took it to the tutor, who was a good scholar, and a very respectable character. At this the dean was exceedingly wroth, and had Mr. Thurlow convened before the Masters and Fellows to answer for his conduct. Thurlow was asked what he had to say for himself. He coolly, perhaps improperly replied, "that what he had done proceeded not from disrespect, but from a feeling of tenderness for the dean; he did not wish to puzzle him!" The dean, greatly irritated, ordered him out of the room; and then insisted that the Masters and Fellows ought immediately to expel or rusticate him. This request was nearly complied with, when two of the Fellows, wiser than the rest, observed, that expelling or rusticating a young man for such an offence would perhaps do much injury to the college, and expose it to ridicule; and that as he would soon quit the college of his own accord to attend the Temple, it would be better to let the matter rest, than irritate him by so severe a proceeding. This advice was at length adopted.

Such was the consciousness which Thurlow felt of his towering abilities, that long before he was called to the bar, he often declared to his friends that he would one day be Chancellor of England; and that the title he would take for his peerage, would be Lord Thurlow, of Thurlow.

Machine for crossing Rivers.—The mechanist, Xavier Michel, residing at Offenbach, has invented a very simple and compact machine, by the aid of which rivers may be crossed, and even the sea attempted, without any danger of sinking. It is nearly five feet in diameter, when unfolded. An opening of about thirteen inches in the centre is destined to receive the traveller. When dismounted, this apparatus is easily transported from place to place—for its entire weight scarcely exceeds five pounds. This inventor has made a number of experiments on the Rhine, all of which have been crowned with entire success. He can make the machine move forward, or

otherwise, at pleasure, and without any great exertion. In order more fully to prove the utility of his invention, M. Michel has determined to embark at Khel, and descend the Rhine to its mouth.

Cattle Scenting Rain.—Liable to long and parching droughts, the author of "*Letters from Buenos Ayres, Chili,*" &c. notices the well-known instinct of cattle in scenting water at a wonderful distance, and describes an occasion wherein it was displayed on the approach of rain, in a similar manner as if a river or spring had been found.

"The negroes were sent in different directions to see how far the scorched grass extended, and were at a considerable distance when the Father Provincial cried out, 'Look at the oxen, they smell water:' we all eagerly turned to the poor panting animals, and saw them stretch out their necks and raise their heads towards the west, and snuff the air in a manner as if they would be certain of obtaining drink could they but raise themselves in the air. At that moment not a cloud nor a single breath of air was to be seen or felt; but in a few minutes the cattle began to move about as if mad, or possessed by some invisible spirit, snuffing the air with most violent eagerness, and gathering closer and closer to each other; and before we could form any rational conjecture as to what could occasion their simultaneous motion, the most tremendous storm came on of thunder, lightning, and rain, I ever witnessed in my life. The rain fell in perpendicular streams as if all the fountains of heaven were suddenly broke loose; so that, in the space of a very few minutes, torrents of water rolled around us, and the cattle easily drank their fill at the spot on which they stood."—*Literary Gazette*.

Irish Bulls.—The secretary of a celebrated Agricultural Society in England, some years ago, in his rage for improvement, and not being overburdened with understanding, sent an order to a bookseller for Mr. and Miss Edgeworth's essay upon *Irish Bulls*, for the use of their society, to assist the members in improving the breed of cattle.

Modern Inventions.—The improvements made in all arts and sciences within the last 200 years have nearly doubled the present limitation of life, in that we live more in less time.

The Egyptians were so ignorant of medicine, that, when any one was sick, they called in as many persons as possible to see him, that, if any one

of them had the like distemper, he might say what was fit for his cure.

Surgery was much the oldest branch of physick which they practised.—

Æsculapius was followed by a dog and a she-goat. The dog was taught to lick all ulcerated wounds, and the goat's milk was given for all diseases of the stomach and lungs.

Receipt to make Yeast.—Three gallons water, two quarts loose hops, boiled together about three hours in brass or bell metal, strain them off from hops, and at once stir in a quart of flour. When cool stir in a pint of good yeast, and half a pound of brown sugar, to remain open in a piggin or jar 15 or 20 hours, and to be stirred often. Put it then in stone jugs about three fourths full, cork them well and place them in a cool situation. Your jugs ought to be of such a size only to contain yeast for the usual quantity of bread baked at a time. One gill of yeast is sufficient for a common sized loaf of bread, that is made from a plate full of flour.

Boots without Seams.—A patent has lately been obtained for the manufacture of boots without seams.—For this purpose, the patentee proposes that the thigh of the beast should be flayed without cutting open, and afterwards dressed and curried upon blocks. The boot top upon the same principle is to be made of the shoulder, prepared in like manner.

FOR THE RURAL MAGAZINE.

The stanzas inserted below are ascribed, it is believed with correctness, to the pen of the late Governor HOWELL, of New Jersey. The occasion which gave rise to them, was that of his leaving home to join the army which was embodied to suppress the Western Insurrection. They breathe a delicacy of feeling, and a warmth of attachment, alike honourable to the author as a husband and a man. They likewise illustrate, very forcibly, the true character of WAR, whose approach is witnessed with sadness and sorrow, by all the endearing charities of domestic life.

I.

THE SOLDIER'S ADIEU.

Ah! Clara, cease—those silent tears
Steal down thy cheeks in vain;
Kind hope shall chase away thy fears,
Till I return again:
But wheresoe'er our route shall be,
My heart shall still encamp with thee.

Why should we lose the single hour,
Which time accords to love,
Suppress that sigh, I own its power,
Yet joy from hope improve:
But wheresoe'er our route shall be,
My heart shall still encamp with thee.

Let no foreboding fears alarm
That regulated mind,
Thy innocence shall shield from harm
Thy soldier far disjoined:
But wheresoe'er our route shall be,
My heart shall still encamp with thee.

Let idle tales of fancied wo,
Ne'er wake for me a fear,
Since honour calls, prepared I go,
Yet dread that parting tear:
But wheresoe'er our route shall be,
My heart shall still encamp with thee.

Start not my fair!—that morning gun
Proclaims 'tis dawn of day,
And now the Reveille's begun,
To hail the morning grey:
But wheresoe'er our route shall be,
My heart shall still encamp with thee.

The general-hark! Oh the adieu!
Permit a last embrace,

~~~~~

When the troops they march, and I'll pursue,  
Farewell that angel face:  
But wheresoe'er our route shall be,  
My heart shall still encamp with thee.

---

FOR THE RURAL MAGAZINE.

## EVENING.

The dusky shadows from the east that rise,  
Steal midway o'er the heavens; the blazing car  
Of Day is sunk; and Sunset's gorgeous dyes  
Fade fast away. Eve's solitary star,  
Watching their golden pomp with kindling eye,  
New trims her virgin lamp: th' unruffled tide  
Gives back a liquid light, while shadows lie  
Deep, broad and strong, the wood-crown'd shores beside.  
How beautiful! all earthly passions fly  
This consecrated hour. The distant bird  
In some sequestered wild mourns on; the fire-fly  
Lights her nuptial torch; the sounds that stirr'd  
Die one by one away. An hour like this  
Is balm unto the soul, steeps every sense in bliss.

Ω

---

### TO THE EDITORS OF THE RURAL MAGAZINE.

The following is taken from a manuscript book of rare pieces, which I have been collecting for upwards of forty years. C. E.

***On the Return of the New Year.***

God's vast existence ne'er decays,  
His age doth never grow,  
Past, present, future, in his sight,  
Are one eternal NOW.

Man measures out his fleeting state  
By motions in the skies,  
And like his own frail vesture, wears  
With every hour that flies.

Successive moments make one day,  
Successive days one year;  
The moments past shall ne'er return,  
Though seasons like appear.

Still a new spring shall bless the earth,  
And a new harvest rise,  
But the last year shall ne'er again  
Revisit mortal eyes.

Old Time, with his keen-pointed scythe,  
Consumes the life of man;  
Our period's less'ning from the hour  
Our beings first began.

Each year fulfills some new event  
Heaven long decreed before,  
Removes unnumbered lives away,  
And gives unnumber'd more.

Soon shall the appointed angel stand  
O'er earth, and air, and sea,  
And swear by him that ever lives,  
That time no more shall be.

Then shall the league of nature cease,  
The sun forsake his way,  
And years and ages lose their name



And years and ages lose their name,  
In one eternal day.

---

# BANK NOTE EXCHANGE,

AT PHILADELPHIA—*June 27th, 1820.*

|                                                                                                                                                                      | Per cent<br>Disc't. |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| U. S. BRANCH BANK Notes,                                                                                                                                             | $\frac{1}{2}$       |
| RHODE ISLAND—generally,                                                                                                                                              | 1                   |
| CONNECTICUT—generally,                                                                                                                                               | 2                   |
| MASSACHUSETTS—Boston,                                                                                                                                                | 1                   |
| Country generally,                                                                                                                                                   | 4-6                 |
| NEW YORK—City Bank Notes,                                                                                                                                            | par.                |
| Country generally,                                                                                                                                                   | 2-5                 |
| NEW JERSEY—generally,                                                                                                                                                | par.                |
| Patterson Bank and Sussex Bank,                                                                                                                                      | 1                   |
| PENNSYLVANIA—Farmer's Bank, of Lancaster; Easton; Delaware<br>County, at Chester; Chester County, at Westchester; Farmer's<br>Bank, Buck's County; Montgomery County | par.                |
| Northampton; Susquehanna Bridge Company; York Bank,<br>Chambersburg,                                                                                                 | $2\frac{1}{2}$ -3   |
| Northumberland; Union,                                                                                                                                               | 20                  |
| Greensburg; Brownsville,                                                                                                                                             | $12\frac{1}{2}$     |
| Farmers & Mechanics' Bank at Pittsburg,                                                                                                                              | 40                  |
| New Hope Bridge Co.                                                                                                                                                  | 1                   |
| DELAWARE—generally,                                                                                                                                                  | par.                |
| Commercial Bank of Delaware; Branch of ditto at Milford,                                                                                                             | 5                   |
| Laurel Bank,                                                                                                                                                         | 50                  |
| MARYLAND—Baltimore Banks,                                                                                                                                            | $\frac{1}{2}$       |
| Baltimore City Bank,                                                                                                                                                 | 5                   |
| Annapolis; Hagerstown,                                                                                                                                               | 2-3                 |
| Snowhill,                                                                                                                                                            | 50                  |

|                                                     |        |
|-----------------------------------------------------|--------|
| Havre de Grace,                                     | 1½     |
| VIRGINIA—Richmond and Branches,                     | 1½     |
| Country generally,                                  | 2½-3   |
| N. W. Bank, at Wheeling,                            | 10-12½ |
| COLUMBIA DISTRICT—Mech. Bank of Alexandria,         | 5      |
| Country generally,                                  | 1      |
| NORTH CAROLINA—State Bank at Raleigh, and Branches, | 8      |
| Cape Fear; Newbern,                                 | 10     |
| SOUTH CAROLINA—State Banks, generally,              | 3      |
| GEORGIA—State Banks, generally,                     | 8      |
| Augusta Bridge Company,                             | 50     |
| TENNESSEE—few sales at any price.                   |        |
| KENTUCKY—No sales.                                  |        |
| OHIO—Marietta,                                      | 15     |
| Steubenville Bank,                                  | 15-20  |
| Bank of Chillicothe,                                | 3      |
| Country generally,                                  | 20-50  |

## PRICES CURRENT—*June 27, 1820.*

|                        | Per          | D. C.  | D. C.     |       |
|------------------------|--------------|--------|-----------|-------|
| Beef, Philad. Mess,    | <i>bbl.</i>  | 13.00  | <i>to</i> | 13.50 |
| Butter, Fresh          | <i>lb.</i>   | 0.18   | "         | 0.12½ |
| Cotton Yarn, No. 10,   | "            | 0.36   | "         |       |
| Flax, Clean, (scarce)  | "            | 0.16   | "         | 0.19  |
| Flaxseed, Clean,       | <i>hhd.</i>  | 12.50  |           |       |
| Firewood, Hickory,     | <i>cord,</i> | 5.50   | "         | 6.25  |
| Oak,                   | "            | 3.00   | "         | 4.00  |
| Flour—Wheat, super.    | <i>bbl.</i>  | 4.50   | "         | 4.62½ |
| Rye,                   | "            | 2.62   | "         | 2.75  |
| Grain—Wheat, (sales)   | <i>bush.</i> | 0.90   | "         | 0.95  |
| Rye, do.               | "            | 0.55   | "         | 0.60  |
| Corn, Pa. do.          | "            | 0.50   | "         | 0.55  |
| Barley,                |              |        |           |       |
| Oats,                  | "            | 0.37   | "         | 0.40  |
| Hams—Jersey,           | <i>lb.</i>   | 0.10   | "         | 0.13  |
| Virginia, (none)       |              |        |           |       |
| Hemp, Kentucky,        | <i>ton,</i>  | 160.00 |           |       |
| Leather—Sole (demand)  | <i>lb.</i>   | 0.24   | "         | 0.27  |
| Upper, undrs'd.        | <i>side,</i> | 2.00   | "         | 2.50  |
| Molasses,              | <i>gall.</i> | 0.50   | "         | 0.55  |
| Nails, Cut, all sizes, | <i>lb.</i>   | 0.07   | "         | 0.12  |
| Pork, Jersey & Penn.   |              |        |           |       |
| Mess, (plenty)         | <i>bbl.</i>  | 15.00  |           |       |
| Plaster of Paris,      | <i>ton,</i>  | 4.50   |           |       |
| Shingles—Cedar,        | 1000         | 25.00  | "         | 27.00 |
| Cypress,               | "            | 5.00   |           |       |
| Seed Clover,           | <i>bush.</i> | 8.50   | "         | 9.00  |
| Wool—Merino, Clean,    | <i>lb.</i>   | 0.75   |           |       |

|                |   |      |   |      |
|----------------|---|------|---|------|
| Do. in Grease, | " | 0.40 | " | 0.45 |
| Common,        | " | 0.50 |   |      |

## RAIN GUAGE AT PHILADELPHIA.

|                                              |          | In.     | Hun. |
|----------------------------------------------|----------|---------|------|
| May                                          | 26,      | Shower, | 0.54 |
|                                              | 29,      | do.     | 0.04 |
|                                              | 13,      | Rain,   | 1.35 |
| June                                         | 6,       | Shower, | 0.19 |
|                                              | 11 & 12, | Rain,   | 0.32 |
|                                              | 13,      | do.     | 0.35 |
|                                              | 16,      | Shower, | 0.04 |
|                                              | 25,      | do.     | 0.30 |
| Total Rain from 17th Jan. to 31st May, 15.45 |          |         |      |

## STATE OF THE THERMOMETER

|         | 9 o'clock. | 12 o'clock. | 3 o'clock. |
|---------|------------|-------------|------------|
| May 27, | —          | 64          | 63         |
| 29,     | 65         | 72          | 73         |
| 30,     | 70         | 76          | 78         |
| 31,     | 63         | 59          | 56         |
| June 1, | 59         | 65          | 68         |
| 2,      | 63         | 70          | 71         |
| 3,      | 68         | 69          | 70         |
| 5,      | 74         | 82          | 83         |
| 6,      | 65         | 71          | 73         |
| 7,      | 67         | 72          | 75         |
| 8,      | 69         | 76          | 80         |
| 9,      | 75         | 84          | 83         |
| 10,     | 77         | 84          | 84         |
| 12,     | 60         | 61          | 63         |
| 13,     | 68         | 75          | 75         |
| 14,     | 75         | 81          | 80         |
| 15,     | 76         | 81          | 80         |
| 16,     | 76         | 82          | 81         |
| 17,     | 71         | 77          | 76         |
| 19,     | 70         | 73          | 76         |
| 20,     | 73         | 77          | 81         |
| 21,     | 75         | 80          | 83         |
| 22,     | 78         | 83          | 86         |
| 23,     | 79         | 83          | 83         |
| 24,     | 79         | 83          | 85         |
| 26,     | 72         | 73          | —          |

---

PHILADELPHIA,  
PUBLISHED MONTHLY BY  
RICHARDS & CALEB JOHNSON,  
*No. 31, Market Street,*  
At \$3.00 per annum.

---

GRIGGS & DICKENSON, *Printers—Whitehall.*

---

## FOOTNOTES:

[1] 1 Cor. chap. 7.

[2] "Theory of Moral Sentiments," by ADAM SMITH, author of the "Wealth of Nations."

[3] A short time before the decease of this very respectable agriculturalist, I had the pleasure of a visit from him, I believe in the year 1815; when I remarked, I had heard much about his celebrated vine from other people, and I now wished some account of it from himself. He informed me, (and I put his statement in writing) that the year previous, he had taken from this vine 40 bushels of grapes, which weighed 2000*lbs.*, from which was made upwards of 100 gallons of wine, the pure juice of the grape; without either water or sugar, or any admixture whatever. The vine and its branches covered the eighteenth part of an acre. Now according to our friend C. E.'s mode of calculation, which is rather too sanguine for practical men, if an eighteenth part produce 100 gallons, a whole acre of course would produce 1800 gallons. It continues to flourish, and bore last year the usual quantity of grapes. Though this is an extraordinary vine, and has received more attention than could be given to any large number by a common sized family, yet it evidently shows the cultivation of this description of vine might be made productive.

Every farmer in the middle and southern states might, if he chose, have such a vine; or at least ten or more smaller ones, which would yield as much, and without any material expense. If this were the case, wine would be so plentiful and so cheap, that every labouring man might have it as a pleasant, cheering and invigorating beverage, and would do more to extinguish the hateful vice of drunkenness than perhaps any other agent within our control.

This sentiment is corroborated by the fact, that in the vine countries of France, where weak wines are as abundant as cider, in a plentiful season in Pennsylvania, and where all, poor and rich, drink of them freely, there is comparatively no



drunkenness. The writer of this note, some years ago, travelled in France in different directions about 1200 miles; and took notice in the whole journey of but two drunken men, and excepting three or four instances, always had his accommodations at an inn, the most likely place to find intemperance. As you go northward into the colder countries of Holland and the north of Germany, where the vine cannot be cultivated but with great difficulty, and wine is too high priced to be commonly used, you may observe the progress of drunkenness almost by the degrees of latitude. Immediately previous to this journey in France, I spent several months in Germany, where I drank coffee regularly twice a day, and was afflicted almost daily with headach. In France, where I seldom used coffee, but frequently weak wine at breakfast, as well as at other times, I had no headach. Should this be generally the effect, it would be another reason in favour of cultivating the vine in the United States. I am no friend to wine bibbers, nor would I be willing to encourage in the remotest degree the use of inebriating liquors, but I should like very much to see a vine such as Joseph Cooper's on every farm in our country.—ED.

[4] The Editors will direct where it may be seen.

[5] Malt, previous to being ground, should be passed through a screen, or sieved to remove the dust.

[6] The smoke pipe, with an upright elbow, about 3 or 4 feet, must be placed on the projecting neck of the fire-place, and with a return elbow convey the smoke through a hole, cut in the brick flue to receive it; by this method the fire will draw well.

If any smoke should come from between the boiler and fire-place, a little dry sand being dropped into the cavity will prevent it.—When the brewing is over, take off the smoke pipe and shake out the soot, which will ensure the fire drawing lively the next brewing.

[7] If the machine is large, the perforated cylinder has four handles for the purpose of easy taking it out and in by a pulley and rope suspended over its centre at a proper height.

[8] If the temperature of the weather is below 55 degrees of heat by the thermometer, it will be better to place your fermenting vessel in a situation not exposed to the cold; the cellar where you keep your beer in would most likely be a good and handy place for this purpose.

[9] If the cask intended for the ale, should not be full, fill it up from your table beer, or if more than enough, put the remainder to the table beer; but this mixing you must regulate according to the strength you want your different sorts of beer.

[10] NOTE.—A person who experienced its benefit and almost certainty, informs us, that he always practised looking steadily into the vapours of the brewing kettle, after the liquor (water) had been in a boiling state for some time. The moment he could distinguish the features of his face, in the surface of the water, he directed the cock to be turned; and the liquor, of course, thrown over the mash. This was an unerring substitute for a thermometer, or sacherometer. His kettle, which had been a still, held about sixty gallons.

[11] Peter Oliver, Esq. then a Judge of the Superior Court of Massachusetts.

[12] The repetition of tillage here reprobated, refers, I presume, to the numerous ploughing given by many English farmers, at that period, preparatory to the putting in of their crops; which the single, deep and "efficacious" ploughing of Arbuthnot rendered unnecessary.—Were our ploughing for Indian Corn and Root Crops alike *deep* and *efficacious, before planting*, shallow tillage (called horse hoeing) with light ploughs, during their growth, would suffice.

[13] In Scotland their peat-lands are called peat-mosses.

[14] The Trench-Plough of Mr. Duckett's invention was so admirably contrived as completely to bury whatever was intended to be turned in. Mr. Young says he saw him turn down a crop of rye, six feet high, so that not an atom was left visible; and yet the depth did not exceed eight inches. Trench ploughing has sometimes been effected in this country by a second plough following in the same furrow after the first, and going a few inches deeper.

[15] This essay, subscribed "A New-England-Man," is published in the 2d volume of the Memoirs of the Philadelphia Society of Agriculture.

[16] Blancourt.

---

#### Transcriber's note:

Minor typographical and punctuation errors have been corrected without note. Irregularities and inconsistencies in the text have been retained as printed.

Mismatched quotes are not fixed if it's not sufficiently clear where the missing quote should be placed.

The cover for the eBook version of this book was created by the transcriber and is placed in the public domain.

Page 280: "Bank of Chillicothe, 3"—The number 3 is unclear.

\*\*\* END OF THE PROJECT GUTENBERG EBOOK THE RURAL  
MAGAZINE, AND LITERARY EVENING FIRE-SIDE, VOL. 1 NO. 07  
(1820) \*\*\*

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

# THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at [www.gutenberg.org/license](http://www.gutenberg.org/license).

## **Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works**

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at [www.gutenberg.org](http://www.gutenberg.org). If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the

work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website ([www.gutenberg.org](http://www.gutenberg.org)), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, “Information about donations to the Project Gutenberg Literary Archive Foundation.”
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.



1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

## **Section 2. Information about the Mission of Project Gutenberg**

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for

generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at [www.gutenberg.org](http://www.gutenberg.org).

### **Section 3. Information about the Project Gutenberg Literary Archive Foundation**

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at [www.gutenberg.org/contact](http://www.gutenberg.org/contact)

### **Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation**

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these

requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit [www.gutenberg.org/donate](http://www.gutenberg.org/donate).

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: [www.gutenberg.org/donate](http://www.gutenberg.org/donate).

## **Section 5. General Information About Project Gutenberg electronic works**

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility: [www.gutenberg.org](http://www.gutenberg.org).

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation,

how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.